

Forward KL Can Be More Robust: Cross-Regularization Loss for Normalizing Flow Boltzmann Sampling

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Abstract

Training normalizing flow Boltzmann generators with the forward KL fails in three complementary ways: it is silent about modes the flow misses (the effective sample size can stay near one while a large fraction of the target mass is absent), it cannot see off-support leakage in regions the target excludes, and it suffers transversal oscillation on thin low-dimensional manifold supports. We address all three by regularizing the forward KL with a family of X functionals X_ω that penalize the pairwise variation of $\log(\mu/\nu)$ under a non-negative weight ω . Three concrete instances are studied, X_μ for in-mode shape control, $X_{\hat{\mu}}$ for stable mode discovery against a wide-coverage measure $\hat{\mu}$, and $X_{\bar{\nu}}$ for off-support leakage suppression, together with a balanced mixture that combines $\hat{\mu}$ and $\bar{\nu}$ inside a single pairwise integral. A Fisher–Rao analysis shows that the forward KL regularized with X_μ decays exponentially with no structural assumption on μ . A second result shows the asymptotic accuracy floor under biased AIS samples is lowered (strictly, unless μ/ν^* is constant on the support of the AIS surrogate). On low-dimensional manifold targets with roughly uniform along-manifold density the framework performs reliably but yields little ESS gain. Heavy-tailed manifold targets remain hard: the quench and temper algorithm that supplies $\hat{\mu}$ tends to miss the thin outer tails, and the regularizer inherits the blind spot.

1 Introduction

Drawing samples from a target distribution $\mu \propto \exp(-U)$ on $\mathbb{R}^d$ known only up to its normalization constant arises across molecular simulation, Bayesian inference, and statistical physics, and is widely tackled with Boltzmann generators built on normalizing flows (Rezende and Mohamed, 2015; Papamakarios et al., 2017; Kingma and Dhariwal, 2018; Noé et al., 2019; Wu et al., 2020). A flow is an invertible parametric map $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ trained so that the pushforward $\nu := (G^{-1})_\# \mu_0$ of an easily samplable reference μ_0 (typically a Gaussian) approximates μ ; by convention G maps the target to the source, $G_\# \mu \approx \mu_0$, so that its inverse G^{-1} maps the source to the target and samples are drawn by pushing μ_0 through G^{-1} . The trained flow returns independent samples from ν along with an exact, tractable density, and importance reweighting against μ then converts those into Monte Carlo estimates with controllable finite-sample error.

Two training objectives dominate the literature. The original Boltzmann generator (Noé et al., 2019) and its variational-inference predecessor (Rezende and Mohamed, 2015) minimize the reverse KL

$$\text{KL}(\nu \parallel \mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx, \quad (1)$$

which needs only samples from ν and is therefore cheap to estimate. It is, however, mode-seeking: on multimodal μ the flow tends to collapse onto a subset of the modes from which it cannot recover (Noé et al., 2019; Wu et al., 2020). The opposing family minimizes the forward KL (Gabrié et al., 2022; Wirmsberger et al., 2020),

$$\text{KL}(\mu \parallel \nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (2)$$

which is mode-covering but needs samples from μ . Such samples are produced by annealed importance sampling (AIS) (Neal, 2001), which transports ν toward μ through a ladder of intermediate distributions while accumulating importance weights. The flow AIS bootstrap (FAB) of Midgley et al. (2022) couples AIS to flow training and replaces the target by the mode-sensitive surrogate μ^2/ν to encourage exploration.

A separate line of work drops the explicit-density requirement and trains a diffusion- or SDE-based sampler directly: the path-integral sampler (Zhang and Chen, 2022), denoising diffusion samplers (Vargas et al., 2023), and the more recent iterated denoising / energy-matching family (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024) learn a stochastic process whose terminal distribution matches μ . These samplers are expressive and scale well, but they do not deliver an exact tractable density of their output, so importance reweighting against μ is no longer available. We commit to flow-based methods for that reason: the invertibility of G gives an explicit ν and therefore an exact, finite-sample importance-sampling pipeline, in which the effective sample size (ESS) (Liu, 2001; Doucet et al., 2001; Martino et al., 2017) computed from the flow’s own weights is a standard and convenient diagnostic of training quality, with known blind spots that motivate the failure-mode analysis below.

Forward KL training has three complementary failure modes that we use to organize the rest of the paper. The first is silent missed modes, the fake ESS pitfall: ESS is gold-standard only when μ and ν are evaluated on the same support, so if ν misses modes of μ , the AIS chain initialized at ν rarely escapes the basin ν already covers (Noé et al., 2019; Midgley et al., 2022) and the resulting ESS can be close to 1 while a large fraction of the mass of μ is missing. The second is off-support leakage. Where μ vanishes, the forward KL integrand $\mu \log(\mu/\nu)$ also vanishes, so the gradient is blind to mass that ν places outside the support of μ . The third is transversal oscillation on low-dimensional manifolds. When μ concentrates on a thin manifold, the forward KL controls only the in-manifold marginal of $\log(\mu/\nu)$, and the off-manifold profile of ν is left free to oscillate. All three blind spots persist no matter how accurately AIS samples the modes ν already reaches.

Our remedy is to regularize the forward KL with a family of X functionals X_ω , named for the pairwise cross-difference of the log-ratio $\log(\mu/\nu)$ that they penalize directly. Let

$$X_\omega(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x) \omega(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (3)$$

where ω is a non-negative density on $\mathbb{R}^d$ (not necessarily normalized), against which both x, y are integrated. The motivation is direct: matching ν to μ is the same as flattening $\log(\mu/\nu)$, and the pairwise difference cancels the unknown normalization of μ . Suitably localized choices of ω provide the three safeguards corresponding to the three failure modes above: stable mode discovery (against silent missed modes), off-support leakage suppression (against gradient blindness off the support of μ), and shape accuracy on low-dimensional manifolds (against transversal oscillation), realized by $X_{\hat{\mu}}$, $X_{\bar{\nu}}$, and X_μ respectively. As a loss ingredient X_ω has three properties that we use throughout. First, it gives pointwise control: the forward KL constrains $\nu \approx \mu$ only in a global Jensen-averaged sense, whereas X_ω penalizes log-ratio variation wherever ω places mass and localizes the constraint to the support of ω . Second, it is robust to the sampling of ω : X_ω vanishes whenever $\log(\mu/\nu)$ is constant on the support of ω , so its minimizer is insensitive to the precise shape of ω , and even hand-injected mode locations can be added to ω without corrupting it. Third, it is compatible with the forward KL: for $\omega = \mu$ the integrand of X_ω coincides with that of the forward KL up to a sign, so the same autograd graph and the same μ -samples are reused at negligible extra cost, the scales match, and no extra hyperparameter is needed.

Each choice of ω localizes the constraint to a region of $\mathbb{R}^d$; we use three choices with complementary roles. With $\omega = \mu$,

$$X_\mu(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (4)$$

which is computed from the same μ -batch as the forward KL at the cost of one extra permutation, by the compatibility above. The forward KL controls the μ -mean of $\log(\mu/\nu)$, whereas X_μ controls its μ -spread, tightening the in-mode density match once ν has reached the modes of μ and reducing the residual bias inherited from imperfect AIS samples.

With $\omega = \hat{\mu}$,

$$X_{\hat{\mu}}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \hat{\mu}(x) \hat{\mu}(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (5)$$

where $\hat{\mu}$ is a coarse, deliberately over-spread approximation of μ . By sample robustness the minimizer of $X_{\hat{\mu}}$ is insensitive to the precise shape of $\hat{\mu}$, so the sole purpose of $\hat{\mu}$ is to place mass on modes and tails that the current ν has not yet covered; the gradient then pulls the flow toward those regions. We build $\hat{\mu}$ once, before training, with a quench and temper (QT) algorithm that diffuses an initial pool through Brownian noise to bridge basins, runs a deterministic mode-finder to collapse it onto mode centers, and rejuvenates with a short Langevin chain. QT errs on the side of broad coverage and works well on targets with disconnected modes and on thin low-dimensional manifolds whose density along the manifold is roughly uniform. It has a known weakness on heavy-tailed targets: the thin, low-mass outer tail is easily missed because the diffusion step rarely scatters particles close enough, and the deterministic mode-finder pulls any near miss back into the wider basins of the body.

With $\omega = \bar{\nu}$,

$$X_{\bar{\nu}}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \bar{\nu}(x)\bar{\nu}(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (6)$$

where $\bar{\nu}$ denotes a stop-gradient copy of the current pushforward ν . Sampling from $\bar{\nu}$ is free, since the training loop already pushes a batch from μ_0 through G^{-1} at every step; the detached gradient keeps the loss numerically stable on sharp targets where a true reverse KL gradient explodes. $X_{\bar{\nu}}$ plays the reverse KL role without its pathologies: it penalizes mass that ν places outside the support of μ , suppressing the mode-overfitting leakage that X_{μ} and $X_{\hat{\mu}}$ alone cannot see.

The three weights are complementary: X_{μ} refines what AIS already reaches, $X_{\hat{\mu}}$ pulls ν toward modes it has missed, and $X_{\bar{\nu}}$ pushes ν away from regions outside the support of μ . Our preferred training loss combines all three with the forward KL,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_{\mu}(\mu\|\nu) + X_{\alpha\hat{\mu}+\beta\bar{\nu}}(\mu\|\nu), \quad (7)$$

where $\lambda, \alpha, \beta \geq 0$ are tunable weights and the third term is X_{ω} with mixture weight $\omega = \alpha\hat{\mu} + \beta\bar{\nu}$. This mixture form goes beyond the self-terms $\alpha^2 X_{\hat{\mu}} + \beta^2 X_{\bar{\nu}}$: expanding the squared base measure, the pairwise integral also generates a cross term of weight $2\alpha\beta$, contributions $|\log(\mu/\nu)(x) - \log(\mu/\nu)(y)|$ with $x \sim \hat{\mu}$ and $y \sim \bar{\nu}$ that anchor the in-support and out-of-support log-ratios to each other.

Equation (7) is written in the abstract density ν , but the trainable object is the flow map G . By the change of variables $\nu(y) = \mu_0(G(y))|\det J_G(y)|$, and writing $U_0 = -\log \mu_0$ for the source potential, the log-ratio along the flow is the single field

$$z(y) := \log \frac{\mu(y)}{\nu(y)} = U_0(G(y)) - U(y) - \log |\det J_G(y)| + \text{const}, \quad (8)$$

up to the additive constant $\log Z_{\mu}$, the unknown log-normalizer of μ , which cancels in every pairwise difference $z(y) - z(y')$ and shifts the forward KL by a constant. Substituting into (7) expresses the preferred loss directly as an objective in G ,

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu} [z(y)]}_{\text{KL}(\mu\|\nu)} + \underbrace{\lambda \mathbb{E}_{y, y' \sim \mu} [z(y) - z(y')]}_{X_{\mu}(\mu\|\nu)} + \underbrace{\mathbb{E}_{y, y' \sim \alpha\hat{\mu}+\beta\bar{\nu}} [z(y) - z(y')]}_{X_{\alpha\hat{\mu}+\beta\bar{\nu}}(\mu\|\nu)}, \quad (9)$$

which is the loss we minimize by stochastic gradient descent and the central object of this paper. Every term is read off a single forward pass of G on a batch, so the regularizers cost essentially nothing beyond a vanilla forward KL step. Section 2 analyzes the Fisher–Rao gradient flow of this loss, and Section 3 details the annealed importance sampling and quench and temper sampling that supply its three expectations.

We pair ESS with the coverage metric of [Naeem et al. \(2020\)](#) for evaluation, which counts the fraction of QT-discovered $\hat{\mu}$ samples whose κ -nearest-neighbor ball is reached by the trained ν . Coverage exposes the modes that a fake ESS hides, so the two metrics together certify both density calibration and mode coverage.

This paper makes four contributions. We introduce the X-functional framework X_{ω} , study three concrete instances X_{μ} , $X_{\hat{\mu}}$, $X_{\bar{\nu}}$, and identify the mixture variant $X_{\alpha\hat{\mu}+\beta\bar{\nu}}$ that combines mode discovery and leakage suppression inside a single pairwise integral. We carry out a Fisher–Rao convergence analysis showing that the forward KL regularized with X_{μ} decays exponentially with no structural assumption on μ , and that the asymptotic accuracy floor under biased AIS samples is lowered by the X_{μ} term (strictly, unless μ/ν^* is constant on the support of the AIS surrogate). We report an empirical study on six 2D benchmarks

(Threewell, Himmelblau, Annulus, Python, Chessboard, Sparse), where, among the four losses we compare, the mixture variant is the only one that both matches or beats the best full-coverage ESS and reaches full mode coverage on every target, together with a high-dimensional product multi-well family up to $d = 256$, where the gap between the X-regularized losses and the bare forward KL widens with the dimension. Finally, we give an adaptive-temperature Boltzmann-generator training scheme (Section 4) that grows a temperature ladder automatically by sizing each step from the classical SMC effective sample size. On a 6-state clock lattice at $D = 64$ the adaptive ladder completes for the X-regularized and the bare forward KL loss alike, along identical ladders; the X terms deliver consistently higher per-stage validation ESS, largest on the early bridges where the training signal rather than the sampling physics is the binding constraint, at no extra cost. The gain at scale is robustness and per-stage quality, a larger training limit rather than a faster transient.

2 Fisher–Rao gradient flow with X functionals

Before turning to the training algorithm, we study the gradient-flow properties of the loss on the space of densities; the convergence rate of this continuous-time idealization is a clean proxy for how trainable the corresponding discrete loss is in practice. We view ν as a generic probability density evolving along the gradient flow of the loss under an appropriate Riemannian metric on the space of densities (Ambrosio et al., 2008; Villani, 2009) and ask how fast ν_t converges to the target μ in $\text{KL}(\mu\|\nu_t)$.

In this section we restrict to the loss with only the forward KL and the shape regularizer X_μ ,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (10)$$

where $\lambda > 0$ is a fixed constant. Throughout this section we abbreviate the importance weight by

$$w(x) := \frac{\mu(x)}{\nu(x)},$$

so that the log-ratio is $z(x) = \log w(x) = \log \mu(x) - \log \nu(x)$, the field z of (8). The importance weight w is distinct from the localization measure ω in X_ω .

The choice of Riemannian metric matters. In the Wasserstein-2 geometry, induced by optimal transport (Jordan et al., 1998; Otto, 2001), the forward KL fails to be displacement convex along Wasserstein-2 geodesics, and the standard Otto–Villani / Bakry–Émery machinery (Otto and Villani, 2000; Bakry et al., 2014) that yields exponential convergence for the reverse KL does not apply. We therefore work in the Fisher–Rao geometry (Amari, 2016; Bauer et al., 2016), in which the forward KL admits an unconditional exponential decay rate and the regularizer X_μ contributes a qualitatively new, nonlocal birth–death modulation (Lu et al., 2019): the local update at x is driven by how the importance weight at x ranks against the population of weights under μ .

Denoting the ν -mean by $\langle \cdot \rangle_\nu := \int (\cdot) \nu dx$, the Fisher–Rao gradient of a functional $\mathcal{F}[\nu]$ and the associated dissipation identity along the FR gradient flow $\partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{F}(\nu_t)$ are

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_\nu \right), \quad \frac{d}{dt} \mathcal{F}[\nu_t] = - \int_{\mathbb{R}^d} \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_{\nu_t} \right)^2 \nu_t dx. \quad (11)$$

The mean-subtraction enforces $\int \text{grad}^{\text{FR}} \mathcal{F} = 0$, so the FR flow conserves probability mass automatically.

2.1 Derivation of X_μ -regularized Fisher–Rao gradient flow

The first variation of the forward KL with respect to ν admits the simple form

$$\frac{\delta \text{KL}(\mu\|\nu)}{\delta \nu}(x) = -\frac{\mu(x)}{\nu(x)} = -w(x), \quad (12)$$

so by (11) the FR gradient of the forward KL is

$$\text{grad}^{\text{FR}} [\text{KL}(\mu\|\nu)](x) = \nu(x) - \mu(x). \quad (13)$$

For the X_μ functional, the inner log-ratio difference has first variation

$$\frac{\delta(\log w(x) - \log w(y))}{\delta\nu(x)} = \frac{1}{\nu(y)}\delta(x - y) - \frac{1}{\nu(x)}.$$

By the symmetry of X_μ in x and y ,

$$\begin{aligned} \frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) &= 2 \int_{\mathbb{R}^d} \mu(x)\mu(y) \operatorname{sgn}(w(x) - w(y)) \frac{\delta(\log w(x) - \log w(y))}{\delta\nu(x)} dy \\ &= 2 \int_{\mathbb{R}^d} \mu(x)\mu(y) \operatorname{sgn}(w(x) - w(y)) \left(\frac{1}{\nu(y)}\delta(x - y) - \frac{1}{\nu(x)} \right) dy \\ &= -2 \int_{\mathbb{R}^d} w(x) \operatorname{sgn}(w(x) - w(y)) dy, \end{aligned}$$

where the contribution of $\delta(x - y)$ vanishes because $\operatorname{sgn}(w(x) - w(y)) = 0$ at $y = x$. Hence

$$\frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) = -2w(x)S(x), \quad S(x) := \mathbb{E}_{y \sim \mu} [\operatorname{sgn}(w(x) - w(y))]. \quad (14)$$

We call S the *rank field* of the importance weight under μ : $S(x) > 0$ where $w(x)$ exceeds the μ -median of w , i.e. where ν undercovers a mode of μ ; $S(x) < 0$ where ν overcovers; and $S \equiv 0$ at $\nu = \mu$. By antisymmetry,

$$\mathbb{E}_\mu[S] = \mathbb{E}_{x, y \sim \mu} [\operatorname{sgn}(w(x) - w(y))] = 0,$$

and the FR gradient of X_μ with respect to ν is

$$\operatorname{grad}^{\text{FR}}[X_\mu(\mu\|\nu)](x) = -2\mu(x)S(x). \quad (15)$$

Since the FR gradient is linear in the functional, the FR gradient of $\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$ is the direct sum of (13) and (15):

$$\operatorname{grad}^{\text{FR}}[\mathcal{L}](x) = \mu(x) - \nu(x) + 2\lambda\mu(x)S(x).$$

The regularized Fisher–Rao gradient flow $\partial_t \nu_t = -\operatorname{grad}^{\text{FR}}[\mathcal{L}]$ therefore reads

$$\partial_t \nu_t(x) = \mu(x) - \nu_t(x) + 2\lambda\mu(x)S_t(x), \quad \text{where } S_t(x) := \mathbb{E}_{y \sim \mu} \left[\operatorname{sgn} \left(\frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right) \right]. \quad (16)$$

Equation (16) is a *nonlocal* PDE: its right-hand side at x depends on $S_t(x)$, which in turn requires the entire profile of the importance weight μ/ν_t over the support of μ . Mass is conserved along the flow because

$$\frac{d}{dt} \int_{\mathbb{R}^d} \nu_t(x) dx = 2\lambda \int_{\mathbb{R}^d} \mu(x) S_t(x) dx = 0,$$

and the flow is stationary at $\nu_t = \mu$, where $\mu(x)/\nu_t(x) \equiv 1$ and hence $S_t \equiv 0$. When $\lambda = 0$, (16) reduces to the pure Fisher–Rao gradient flow of the forward KL.

2.2 Convergence analysis of the X_μ -regularized flow

To analyze the convergence rate of the regularized flow (16), we take the forward KL $\text{KL}(\mu\|\nu_t)$ as the error metric and differentiate it along the flow:

$$\frac{d}{dt} \text{KL}(\mu\|\nu_t) = - \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} (1 + 2\lambda S_t) dx + \int_{\mathbb{R}^d} \mu dx = -\chi^2(\mu\|\nu_t) - 2\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx,$$

where the chi-squared divergence is

$$\chi^2(\mu\|\nu) := \int_{\mathbb{R}^d} (w - 1)^2 \nu dx = \int_{\mathbb{R}^d} w^2 \nu dx - 1.$$

The sign of the second integral is settled by the following rank-correlation identity.

Lemma 1 For any probability density ν on $\mathbb{R}^d$,

$$\int_{\mathbb{R}^d} \frac{\mu(x)^2}{\nu(x)} S(x) dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|] \geq 0,$$

with equality if and only if w is μ -almost-everywhere constant, i.e. $\nu = \mu$.

Proof. Since $\mu^2/\nu = \mu w$, the definition of S gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \mathbb{E}_{\mu} [w(x) S(x)] = \mathbb{E}_{x,y \sim \mu} [w(x) \operatorname{sgn}(w(x) - w(y))].$$

Symmetrizing in $x \leftrightarrow y$ and averaging the two equal copies gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [(w(x) - w(y)) \operatorname{sgn}(w(x) - w(y))] = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|].$$

□

Applying Lemma 1 with $\nu = \nu_t$ and $S = S_t$, the dissipation rewrites as

$$\frac{d}{dt} \operatorname{KL}(\mu \| \nu_t) = -\chi^2(\mu \| \nu_t) - \lambda \mathbb{E}_{x,y \sim \mu} \left[\left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right]. \quad (17)$$

The elementary inequality $w \log w \leq w^2 - w$ applied to $w = \mu/\nu_t$ gives $\chi^2(\mu \| \nu_t) \geq \operatorname{KL}(\mu \| \nu_t)$, hence $\frac{d}{dt} \operatorname{KL}(\mu \| \nu_t) \leq -\operatorname{KL}(\mu \| \nu_t)$, and Grönwall's lemma yields the exponential decay

$$\operatorname{KL}(\mu \| \nu_t) \leq e^{-t} \operatorname{KL}(\mu \| \nu_0), \quad t \geq 0, \quad (18)$$

with no structural assumption on μ : neither log-concavity, nor a spectral gap, nor any growth condition. This contrasts sharply with the Wasserstein-2 geometry, where the forward KL is not displacement convex and an analogous exponential rate is unavailable without further assumptions.

The uniform convergence rate (18) should *not* be read as a blanket statement that minimizing the forward KL is intrinsically faster or more efficient than minimizing the reverse KL. The FR flow is a continuous-time idealization on the space of densities and is agnostic to the sampling cost. In practice, the difficulty of training has been partially transported to the AIS step: evaluating the forward KL and its gradient requires accurate samples of μ , and producing such samples in high dimensions is itself a hard problem. This is the very reason that AIS is needed, and that the $X_{\hat{\mu}}$ regularizer must compensate for modes that AIS alone fails to reach. The exponential rate (18) quantifies how fast the flow *can* converge given accurate target samples; it does not eliminate the burden of producing them.

A naive comparison of dissipation rates between KL-only and $\operatorname{KL} + \lambda X_{\mu}$ would also be misleading, since scaling the loss by a constant is mere time reparametrization with no qualitative gain. The substantive contribution of X_{μ} is structural rather than rate-based: while the pure-KL FR gradient is local in x , the rank field S_t is global, so the regularized flow (16) couples the update at x to the entire distribution of importance weights under μ . This nonlocal coupling amplifies the birth rate on undercovered regions of μ and suppresses it on overcovered ones, a global comparison that pure forward KL (which knows only whether $\mu(x)/\nu_t(x)$ exceeds or falls below 1 pointwise) cannot produce.

Two further cautions concern the discrete realization of the flow. First, the time-reparametrization argument cuts in both directions once the optimizer enters. Training follows Adam steps, not the FR flow: the per-coordinate second-moment normalization absorbs a global rescaling of the loss, so a larger gradient buys no speedup by itself, while a regularizer that reshapes the relative magnitudes of the gradient components does change the optimizer's effective time scale. Part of any observed difference in training speed between KL and $\operatorname{KL} + \lambda X_{\mu}$ may therefore be optimizer scaling rather than information geometry, and we claim no faster transient on this basis; the rejected-rung values of Section 5.4 confirm that none materializes. Second, the dissipation (17) is a population statement, while each gradient step sees a B -sample estimate. At large batch the estimated forward KL gradient already approximates its population direction well, the idealized rate (18) is nearly realized by the bare loss, and the headroom left for the X terms narrows; the regularizer earns its keep when the training signal is noisy or biased: small batches, early bridges, imperfect AIS samples. The clock ladder of Table 5 shows this contraction directly, the early-rung validation-ESS gap shrinking from +0.21 at $B = 10^3$ to +0.13 at $B = 10^5$.

Remark (Monotone decay of X_μ) The regularizer X_μ also decreases monotonically along the regularized flow: pairing the variation $\delta X_\mu / \delta \nu = -2wS$ (14) against the flow (16) gives

$$\frac{d}{dt} X_\mu(\mu \| \nu_t) = -2 \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx - 4\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t^2 dx \leq 0,$$

with strict inequality whenever $\nu_t \neq \mu$, the first term being strictly negative by Lemma 1. Thus X_μ is itself driven to zero in the long-time limit.

2.3 Asymptotic accuracy of biased forward KL training

The convergence analysis of Section 2.2 assumes that the outer expectation in $\text{KL}(\mu \| \nu) + \lambda X_\mu(\mu \| \nu)$ is taken under the exact target μ . In practice, the algorithm only has access to an approximation $\tilde{\mu}$ of μ supplied by AIS, and the realized loss reads

$$\tilde{\mathcal{L}}[\nu] = \int_{\mathbb{R}^d} \tilde{\mu}(x) \log w(x) dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) |\log w(x) - \log w(y)| dx dy, \quad (19)$$

where the pointwise ratio $w(x) = \mu(x)/\nu(x)$ is still evaluated against the exact target, since it is computed per sample from the target potential U and the flow's own density. Repeating the derivation of Section 2.1 with μ replaced by $\tilde{\mu}$ in the outer integrands gives the Fisher–Rao gradient

$$\text{grad}^{\text{FR}}[\tilde{\mathcal{L}}](x) = \nu(x) - \tilde{\mu}(x)(1 + 2\lambda \tilde{S}(x)), \quad \tilde{S}(x) := \mathbb{E}_{y \sim \tilde{\mu}} [\text{sgn}(w(x) - w(y))].$$

A stationary point ν^* of (19) therefore satisfies the self-consistent equation

$$\nu^*(x) = \tilde{\mu}(x) \left[1 + 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(y) \text{sgn} \left(\frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right) dy \right]. \quad (20)$$

At $\lambda = 0$ the equation collapses to $\nu^* = \tilde{\mu}$: pure forward KL training inherits the bias of the AIS surrogate $\tilde{\mu}$ verbatim, and the residual error is exactly $\text{KL}(\mu \| \tilde{\mu})$. For $\lambda > 0$, however, the rank-based correction $1 + 2\lambda \tilde{S}^*$, where $\tilde{S}^*$ denotes $\tilde{S}$ evaluated at $\nu = \nu^*$, inflates ν^* above $\tilde{\mu}$ where the importance weight μ/ν^* is high (i.e. where ν^* still undercovers μ) and deflates it where the weight is low, providing a global feedback that pulls ν^* from $\tilde{\mu}$ back toward μ . This is quantified in the following theorem.

Theorem 1 Assume μ is absolutely continuous with respect to $\tilde{\mu}$ and $\lambda \in [0, 1/2]$, so that the bracket in (20) is non-negative; together with $\int \tilde{\mu}(x) \tilde{S}^*(x) dx = \mathbb{E}_{x, y \sim \tilde{\mu}} [\text{sgn}(w(x) - w(y))] = 0$ (antisymmetry in $x \leftrightarrow y$), this makes $\nu^* = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)$ a probability density. Any stationary point ν^* of the biased loss (19) satisfies

$$\text{KL}(\mu \| \nu^*) \leq \text{KL}(\mu \| \tilde{\mu}) - \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy. \quad (21)$$

Proof. Substituting the stationary identity $\nu^* = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)$ into the difference of forward KLs and using $\mu = \nu^* \cdot (\mu/\nu^*) = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)(\mu/\nu^*)$,

$$\text{KL}(\mu \| \tilde{\mu}) - \text{KL}(\mu \| \nu^*) = \int_{\mathbb{R}^d} \mu(x) \log(1 + 2\lambda \tilde{S}^*(x)) dx = \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} u(x) \log u(x) dx,$$

where $u(x) := 1 + 2\lambda \tilde{S}^*(x) \geq 1 - 2\lambda \geq 0$, since $\tilde{S}^* \in [-1, 1]$ (a $\tilde{\mu}$ -average of signs) and $\lambda \leq 1/2$. The elementary inequality $u \log u \geq u - 1$ for $u \geq 0$ then yields

$$\text{KL}(\mu \| \tilde{\mu}) - \text{KL}(\mu \| \nu^*) \geq 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx.$$

Applying the rank-correlation argument of Lemma 1 verbatim with μ replaced by $\tilde{\mu}$ in the outer measure,

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx = \frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy,$$

which gives (21). $\square$

The right-hand side of (21) contrasts the two roles of X_μ training. At $\lambda = 0$ the bound is tight at $\nu^* = \tilde{\mu}$ and the residual KL is fixed at $\text{KL}(\mu \parallel \tilde{\mu})$; the AIS bias is a hard floor. Switching on the regularizer with $\lambda \leq 1/2$ lowers this floor (strictly, unless μ/ν^* is constant on the support of $\tilde{\mu}$) by an amount controlled by the discrepancy of the importance weights μ/ν^* across the support of $\tilde{\mu}$: the larger this discrepancy, the more the rank-based feedback pulls ν^* toward μ . The X_μ regularizer is therefore not merely a numerical stabilizer for the forward KL flow: it directly improves the asymptotic accuracy of the trained density when the target samples are biased, which is the practically relevant regime.

The threshold $\lambda \leq 1/2$ is a worst-case condition enforcing positivity of ν^* uniformly in x ; in practice the accuracy improvement persists well beyond it. The practical takeaway is that even without the wide-coverage regularizer $X_{\tilde{\mu}}$, adding X_μ alone to the forward KL loss already reduces the bias floor inherited from inaccurate AIS samples.

3 Flow training with X functionals

This section turns the abstract loss of the introduction into a concrete training algorithm for the flow map G . Let U_0 and U be the source and target potentials on $\mathbb{R}^d$, so that $\mu_0 \propto \exp(-U_0)$ is the easily samplable reference and $\mu \propto \exp(-U)$ is the target. We parameterize $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ so that $G_{\#}\mu \approx \mu_0$ — mapping target to source, so that no inversion is required during training — and write $\nu := (G^{-1})_{\#}\mu_0$ for the induced approximation to μ . We first assemble the training loss and read off the two sampling ingredients it requires (Section 3.1), then supply them in turn: annealed importance sampling of μ (Section 3.2), the quench and temper construction of the wide-coverage measure $\hat{\mu}$ (Section 3.3), and finally the ESS and coverage diagnostics used to evaluate the trained flow (Section 3.4).

3.1 The X-regularized training loss

By the change-of-variables formula the pullback density of μ_0 under G is $\nu(y) = \mu_0(G(y))|\det J_G(y)|$, so the log-ratio between the target and the model along the flow is the single scalar field $z(y)$ of (8). Both ingredients of the loss are elementary functionals of z . The forward KL is its mean under the target, $\text{KL}(\mu \parallel \nu) = \mathbb{E}_{y \sim \mu}[z(y)] + \text{const}$ (cf. (2)), and each X functional X_ω (3) is the mean absolute pairwise variation of z under a non-negative weight ω , $X_\omega(\mu \parallel \nu) = \mathbb{E}_{y, y' \sim \omega}[|z(y) - z(y')|]$ when ω is normalized (in general it is the double integral (3)). Combining the shape term $\omega = \mu$ with the balanced mixture $\omega = \alpha\hat{\mu} + \beta\bar{\nu}$ that handles mode discovery and leakage suppression, the total training loss in terms of the flow map G is

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu}[z(y)]}_{\text{KL}(\mu \parallel \nu)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu}[|z(y) - z(y')|]}_{X_\mu(\mu \parallel \nu)} + \underbrace{\mathbb{E}_{y, y' \sim \alpha\hat{\mu} + \beta\bar{\nu}}[|z(y) - z(y')|]}_{X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu \parallel \nu)}. \quad (22)$$

We call the objective (22), previewed as (9) in the introduction, the *X-regularized forward KL* (the cross-regularization loss of the title). We stress the two pushforward measures appearing in (22). The model distribution $\nu = (G^{-1})_{\#}\mu_0$, with density $\nu(y) = \mu_0(G(y))|\det J_G(y)|$, is the flow’s live approximation to the target μ and depends on the trainable parameters of G . The barred measure $\bar{\nu}$ is the *same* pushforward frozen at the current parameters (a stop-gradient copy of ν), which enters the mixture term only as a sampling weight and is never differentiated through.

The three sampling measures entering (22) address complementary failure modes of the bare forward KL:

- μ : *accuracy at small batch*. The shape term X_μ compares the importance weights pairwise across the batch and lowers the asymptotic accuracy floor under biased AIS samples (Section 2.3); the gain is largest when the batch is small and the B -sample KL gradient is noisy.
- $\hat{\mu}$: *mode discovery*. The wide-coverage measure built by quench and temper (Section 3.3) places sample pairs across modes that AIS from the current pushforward cannot reach, pulling ν toward mass the bare loss never sees.

- $\bar{\nu}$: *leakage suppression*. The frozen pushforward penalizes the mass the flow places outside the support of μ , preventing the overfit that target samples alone cannot detect: where μ provides no samples, the bare forward KL provides no signal.

None of the three changes the minimizer, each X term vanishing at $\nu = \mu$; what they add to the existing forward KL is numerical robustness, not a new objective.

Remark (Robustness, not speed) *As discussed in Section 2.2, the exponential rate (18) is a statement about the idealized FR flow: the discrete Adam dynamics carries no such guarantee and may take many steps to converge. Moreover, at large batch the margin over the bare forward KL can be minor, since a large batch is already a powerful safeguard against mode collapse in practice; the gap is widest in the small-batch regime (Table 5).*

All three terms are computed in a single forward pass: evaluate $\mathbf{z} = (z(y_1), \dots, z(y_B))$ once on a batch, take $\text{mean}(\mathbf{z})$ for the forward KL, and $\text{mean}(|\mathbf{z} - \mathbf{z}[\sigma]|)$ for a random permutation σ of the batch for each X term. Backpropagation passes through $\mathbf{z}$ exactly once, identical to a vanilla forward KL step, so the X functionals add essentially no extra cost.

The loss draws on three measures, of which only two must be built. The $\bar{\nu}$ batch is free: it is the raw pushforward $\bar{y}_i = G^{-1}(x_i)$ of source samples $x_i \sim \mu_0$, taken with the gradient through G detached (a stop-gradient); this detached sampler keeps the loss numerically stable on sharp targets, where differentiating through the sampling chain (as a true reverse KL gradient does) would explode. The other two require construction: samples of the target μ for the forward KL and X_μ terms, supplied by annealed importance sampling (Section 3.2), and samples of the wide-coverage measure $\hat{\mu}$ for the mixture term, supplied by quench and temper (Section 3.3).

Although (λ, α, β) are three free parameters in principle, the X functional is by construction on the same scale as the forward KL, so any choice of order 1 works in practice. Two settings are worth highlighting. The *X_μ -only hyperparameters* $\lambda = 1, \alpha = \beta = 0$ add only X_μ on top of the forward KL; this term reuses the AIS samples and autograd graph of the forward KL, costing only an extra permutation and elementwise difference per batch, and it suffices whenever the source μ_0 already overlaps every mode of μ . The *balanced hyperparameters* $\lambda = 1, \alpha = \beta = 1/2$ turn on the mixture term, an equal-weight draw from $\hat{\mu}$ and $\bar{\nu}$: the $\hat{\mu}$ half pulls ν toward QT-discovered modes that AIS alone cannot reach, while the $\bar{\nu}$ half restrains the mass the flow places outside the support of μ . We recommend the balanced hyperparameters as the default whenever μ_0 may not cover all modes of μ , and use them throughout the experiments; re-tuning (λ, α, β) away from these defaults did not improve results on any of our benchmarks.

3.2 Sampling μ by annealed importance sampling

Evaluating the μ -expectations in (22) requires samples from μ , which are not directly available. The natural surrogate draws $y_1, \dots, y_B$ from the pushforward $\nu = (G^{-1})_{\#}\mu_0$ and reweights by

$$w(y) = \frac{\mu(y)}{\nu(y)} = \exp(z(y)) = \exp\left(U_0(G(y)) - U(y) - \log |\det J_G(y)|\right), \quad y \in \mathbb{R}^d, \quad (23)$$

followed by a resampling step and a short Langevin simulation to diffuse the duplicates. In high dimensions, however, μ and ν typically differ greatly, so the weights $w(y_i)$ collapse onto a few outliers and the resulting samples are of very low quality. Annealed importance sampling (AIS) bridges this gap by inserting a ladder of $M + 1$ distributions

$$\pi_m(y) \propto \nu(y)^{1-\frac{m}{M}} \mu(y)^{\frac{m}{M}} = ((G^{-1})_{\#}\mu_0)(y) w(y)^{\frac{m}{M}}, \quad m = 0, 1, \dots, M, \quad (24)$$

so that $\pi_0 = \nu$ and $\pi_M = \mu$. Each stage of the ladder reweights by only $w(y)^{\frac{1}{M}}$, keeping the weight ratio close to 1 and avoiding the degeneracy of direct importance sampling.

In particular, when G is the identity map the proposal collapses to the source, $(G^{-1})_{\#}\mu_0 = \mu_0$, the Jacobian term drops out of w , and Algorithm 1 reduces to classical sequential Monte Carlo (SMC) bridging μ_0 to μ .

In the general case, the Langevin step nominally targets the intermediate distribution π_m , and so requires its score $\nabla_y \log \pi_m(y)$. Through the pushforward density $\nu(y) = \mu_0(G(y)) |\det J_G(y)|$, this score carries the

Algorithm 1: Annealed importance sampling (AIS) from $(G^{-1})_{\#}\mu_0$ to μ

Input: samples $y_1, \dots, y_B$ from $(G^{-1})_{\#}\mu_0$, ladder length M , intermediate distributions π_m (24)
Output: samples $y_1, \dots, y_B$ approximately distributed as μ

- 1 **for** $m = 1, \dots, M$ **do**
 - // reweight along the geometric path*
 - 2 Compute the incremental weights on $y_1, \dots, y_B$, $w(y_i)^{\frac{1}{M}} = \pi_m(y_i)/\pi_{m-1}(y_i) = \exp(z(y_i)/M)$;
 - // resample*
 - 3 Resample from the weighted set, $y'_1, \dots, y'_B \leftarrow \text{resample}(y_{1:B}, (w(y_i)^{\frac{1}{M}})_{i=1}^B)$;
 - // rejuvenate under the target*
 - 4 Diffuse the resampled duplicates by a short Langevin chain on μ , $y_{1:B} \leftarrow \text{langevin}(y'_{1:B}, U)$;
- 5 **end**

term $\nabla_y \log |\det J_G(y)|$, whose evaluation needs the Jacobian $J_G(y)$ itself (not merely its determinant, which the flow already provides), and is therefore expensive for a deep map. In practice we instead diffuse the resampled particles directly under the target μ , whose score $\nabla_y \log \mu(y) = -\nabla U(y)$ is cheap. Since the per-stage reweighting already carries the exact π_m correction along the geometric path (24) and μ is the ultimate target, the only effect of the substitution is on intermediate mixing, not on the weighted target: the Langevin kernel serves only to decorrelate the duplicates left by resampling.

Remark (Choosing the ladder length M) For simple targets close to the source, $M = 1$ already suffices. The choice $M = d$ is typical but not necessary in every test — an empirical rule by which the ladder length scales with the dimension of the deformation the flow must bridge, of which d is the worst case. For instance, annealing $\mu_0 \propto e^{-|x|^2}$ to $\mu \propto e^{-2|x|^2}$ on $\mathbb{R}^d$ deforms all d coordinates at once, and a ladder of $M = d$ steps keeps the weighted empirical distribution at a finite ESS that does not degrade with d ; when the deformation concentrates on fewer directions, a much shorter ladder suffices.

3.3 Constructing $\hat{\mu}$ by quench and temper

The mixture term needs samples from the wide-coverage measure $\hat{\mu}$. These are prepared once, before training, as a fixed pool of P samples $x_1, \dots, x_P$ distributed as $\hat{\mu}$; at each training step a mini-batch is drawn from this set and refreshed by a short Langevin chain before entering the X_ω computation of Section 3.1 with μ replaced by $\hat{\mu}$. The set is produced by quench and temper (QT), which proceeds in three stages: diffusion (melt), optimization (quench), and rejuvenation (temper).

Algorithm 2: Quench and temper (QT): proactive mode discovery

Input: target potential U (with fixed pool size P and diffusion scale σ_{QT})
Output: wide-coverage set $\hat{\mathcal{Y}} = \{x_1, \dots, x_P\}$ forming $\hat{\mu}$, written $\hat{\mathcal{Y}} = \text{QT}(U)$

- 1 Draw an initial pool $x_1, \dots, x_P \sim \mu_0$;
- // diffusion (melt)*
- 2 Scatter the samples to a wider region of $\mathbb{R}^d$, $x_i \leftarrow x_i + \sigma_{\text{QT}} \xi_i$ with $\xi_i \sim \mathcal{N}(0, I_d)$ i.i.d. and large $\sigma_{\text{QT}} > 0$;
- // optimization (quench)*
- 3 Drive each sample toward a mode center of μ by L-BFGS on U , updating $x_1, \dots, x_P$;
- // rejuvenation (temper)*
- 4 Spread the samples around each mode, $x_{1:P} \leftarrow \text{langevin}(x_{1:P}, U)$;

The output samples concentrate near the mode centers of μ , and the initial Brownian diffusion ensures that

even modes far from the support of the input are reachable. The $X_{\hat{\mu}}$ regularizer therefore penalizes any mode of μ on which the current pushforward ν places too little mass, driving the flow to discover modes that AIS alone could never reach.

A large diffusion σ_{QT} biases the QT output toward modes that lie far from the bulk of μ_0 . This is desirable rather than problematic: the modes the current pushforward fails to cover are precisely those far from μ_0 , and the bias is also harmless because the minimizer of $X_{\hat{\mu}}$ is insensitive to the precise shape of $\hat{\mu}$.

Remark (QT failure mode: heavy-tailed manifolds) *On heavy-tailed manifold targets the thin outer tail consists of narrow-variance components whose basins are too small for the L-BFGS quench to land in: the diffusion step rarely scatters particles close enough, and any near miss is pulled back into the wide-basin body. The resulting $\hat{\mu}$ traces the body but misses the tail, and $X_{\hat{\mu}}$ inherits the blind spot.*

3.4 Importance sampling at evaluation: ESS and coverage

At evaluation time the trained flow G supplies the model distribution $\nu = (G^{-1})_{\#}\mu_0$ as an approximation to μ , together with the exact tractable density $\nu(y) = \mu_0(G(y))|\det J_G(y)|$ that any normalizing flow makes available. Because the density is exact, the importance weight $w(y) = \mu(y)/\nu(y) = \exp(z(y))$ of (23) can be evaluated for free on any pushforward sample and used at test time to correct the residual bias of ν as an approximation to μ . Drawing $\mathbf{y} = (y_1, \dots, y_N)$ with $y_i = G^{-1}(x_i)$ and $x_i \sim \mu_0$ i.i.d., the weights $w(y_i)$ are put to two uses. First, they diagnose how well ν approximates μ , through the effective sample size defined below. Second, they turn the pushforward into approximate samples of μ : one resamples $\mathbf{y}$ by the weights and refreshes with a short Langevin chain targeting μ , exactly the surrogate that drove the AIS step in Section 3.2 but now applied with the trained flow held fixed. The training bias the X functionals were designed to reduce is, at this stage, traded for sampling cost: the better the flow, the more uniform the weights and the less reweighting the resampling has to do.

Remark (Coarse training, fine importance sampling) *If the target has a natural coarse representation, the flow can be trained on the cheap low-dimensional coarse version and the resulting proposal reweighted by the importance weights against the fine high-dimensional target at evaluation — the high-dimensional target then never enters a gradient step. This separation is the way Lin et al. (2026) applies the framework to path integral Monte Carlo.*

We assess the quality of this pipeline through two complementary metrics: the effective sample size (ESS), which measures how uniform the importance weights are on the support reached by the flow, and the coverage, which checks whether every mode of μ has been reached at all.

The first metric is the effective sample size, normalized to $[0, 1]$ (the classical ESS divided by the sample count N). Let $\mathbf{y} = (y_1, \dots, y_N) \in \mathbb{R}^{N \times d}$ be i.i.d. samples from ν , with importance weights $w(y_i)$ given by (23). The ESS is defined as

$$\text{ESS}(\mathbf{y}) = \frac{(\sum_{i=1}^N w(y_i))^2}{N \sum_{i=1}^N w(y_i)^2} \in [0, 1], \quad (25)$$

where the upper bound 1 follows from the Cauchy–Schwarz inequality and is attained when all weights are equal. When G is perfectly trained the weights are constant and the ESS reaches 1. As emphasized in the introduction, however, this conclusion is only one-way: a high ESS certifies that the weights are uniform *on the support of ν* , but modes of μ that the pushforward never reaches contribute nothing to the sum and so cannot lower the ESS, which gives rise to the *fake ESS* pitfall.

To detect such missing modes, we pair the ESS with the coverage metric of Naeem et al. (2020). Let $\mathbf{x} = (x_1, \dots, x_P) \in \mathbb{R}^{P \times d}$ be the samples of $\hat{\mu}$ produced by the quench and temper algorithm (Algorithm 2). For each $1 \leq i \leq P$, the κ -nearest-neighbor radius of x_i (with $\kappa = 5$) is

$$\text{NND}_{\kappa}(x_i; \mathbf{x}) = \inf \left\{ r \geq 0 : \#\{1 \leq j \leq P : 0 < |x_i - x_j| < r\} \geq \kappa \right\}, \quad (26)$$

i.e., the distance from x_i to its κ -th nearest neighbor in $\mathbf{x}$. The coverage of $\mathbf{x}$ by the pushforward samples $\mathbf{y}$ is then

$$\text{Coverage}_{\kappa}(\mathbf{y}; \mathbf{x}) = \frac{1}{P} \sum_{i=1}^P \mathbf{1} \left\{ \exists 1 \leq j \leq N : |y_j - x_i| \leq \text{NND}_{\kappa}(x_i; \mathbf{x}) \right\}, \quad (27)$$

i.e., the fraction of points in $\mathbf{x}$ whose κ -nearest-neighbor ball contains at least one pushforward sample. A coverage close to 1 certifies that every region populated by $\hat{\mu}$ has been reached by the flow, while a coverage substantially below 1 exposes precisely the missing modes that the ESS alone cannot see. Because $\hat{\mu}$ is the same wide-coverage measure that $X_{\hat{\mu}}$ trains toward, the coverage metric certifies reach of the QT-discovered modes only, and inherits QT’s blind spot for thin heavy tails (Remark 4).

4 Adaptive-temperature Boltzmann generator

Section 3 introduced the X-regularized forward KL (22), which improves the sample quality of the bare forward KL. In high dimensions — molecular potentials being the prime example — training a single flow map directly from the source $\mu_0 \propto \exp(-U_0)$ to the target $\mu \propto \exp(-U)$ is essentially infeasible, however good the loss. The standard remedy in Boltzmann generators is to decompose this one hard problem into a sequence of easier ones along a temperature ladder, so that each flow need only learn a single stage — a small change in the potential. This section shows how to build that ladder adaptively from the ESS metric.

4.1 Construction of temperature ladder

To bridge the simple source $\mu_0 \propto \exp(-U_0)$ to the target $\mu \propto \exp(-U)$, we seek an increasing sequence

$$0 = t_0 < t_1 < \cdots < t_K = 1$$

defining a sequence of intermediate distributions

$$\mu_k \propto \exp(-U_k), \quad U_k(x) = (1 - t_k) U_0(x) + t_k U(x), \quad k = 0, 1, \dots, K,$$

so that μ_0 is the source and μ_K the target, together with K flow maps $\{G_k\}_{k=1}^K$ trained so that $(G_k^{-1})_{\#} \mu_{k-1} = \mu_k$. The number of stages K is left flexible; the crucial design choice is each increment $\Delta t_k = t_k - t_{k-1}$. Too large an increment degrades the training quality of stage k , while too small an increment wastes computation, so we want the largest step that each flow can still learn well.

Training the flow G_k requires samples from its target μ_k , which we obtain by SMC (Algorithm 1 with G the identity). The smallest per-step ESS of that SMC run is then a natural criterion for the step Δt_k : it measures how well the run bridges μ_{k-1} to μ_k , so we enlarge the step until this ESS drops to a fixed floor.

These SMC runs are purely classical (they involve only the potentials U_0 and U , never the flow map) and are therefore cheap. They also use no QT samples, so the per-step ESS they report may be a fake ESS that overlooks modes the SMC chain never reaches. This is harmless here, because Algorithm 3 selects only the increment Δt_k and never produces training samples, so its ESS is used solely to gauge the shape deformation $\mu_{k-1} \rightarrow \mu_k$ and needs no faithful target. The floor $\tau = 0.5$ and shrink factor $\gamma = 0.7$ are empirical; the initial guess t_{init} is supplied by the outer loop (Algorithm 4). The procedure returns the largest admissible step below the initial guess: the coarsest level t_k , and hence the most distant intermediate U_k , on which the flow G_k can still be trained reliably from U_{k-1} . It always terminates, since as $t_k \rightarrow t_{k-1}$ the bridge $\mu_{k-1} \rightarrow \mu_k$ collapses and $\text{ESS}_{\min} \rightarrow 1 \geq \tau$, so the shrink loop is exited after finitely many steps.

4.2 The full training procedure

The two ingredients now assemble into a single loop (Algorithm 4) that grows the ladder one stage at a time, from U_0 to U , carrying a validation set $\mathcal{Y}_k \approx \mu_k$ across stages. Stage k runs five steps, with steps (iii)–(v) wrapped in an acceptance loop that shrinks the increment and retrains whenever the trained flow’s validation ESS falls below a floor τ_v ; the outer loop stops once t_k reaches 1, so the number of stages K is selected automatically.

- (i) *Source samples.* Stage k consumes source samples in three distinct ways, governed by two size parameters: the *pool size* P and the *batch size* B . First, a selection pool of P samples, drawn from the validation set $\mathcal{Y}_{k-1} \approx \mu_{k-1}$ carried over from the previous stage and refreshed by a short Langevin rejuvenation on U_{k-1} , supplies the SMC particles for the adaptive step selection of step (ii). Second,

Algorithm 3: Adaptive selection of the temperature step Δt_k

Input: potentials U_0, U , previous level t_{k-1} , selection pool $\mathcal{P}$ of P samples from μ_{k-1} , floor $\tau = 0.5$, shrink factor $\gamma = 0.7$, initial guess t_{init}

Output: next level t_k , giving the step $\Delta t_k = t_k - t_{k-1}$

```
1  $U_{k-1} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U;$   
   // start from the supplied initial guess  
2  $t_k \leftarrow t_{\text{init}};$   
3 while true do  
   // run classical SMC across the candidate step  
4   Run  $M$ -step SMC on the pool  $\mathcal{P}$  (Algorithm 1 with  $G$  the identity,  $M$  as in Remark 3) from the  
     distributions  $\mu_{k-1} \propto e^{-U_{k-1}}$  to  $\mu_k \propto e^{-U_k}$ , where the potential  $U_k = (1 - t_k)U_0 + t_k U$ ;  
   // score the step by its worst per-step ESS  
5   Record  $\text{ESS}_{\min}$ , the smallest per-step ESS over the  $M$  steps;  
6   if  $\text{ESS}_{\min} \geq \tau$  then  
7     return  $t_k$ ;  
8   else  
9     // shrink the step toward  $t_{k-1}$   
      $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1});$   
10  end  
11 end
```

the quench and temper of step (iii) melts an initial pool of P source samples (Algorithm 2). Third, the training batches of step (iv) are drawn at run time, B samples per gradient step, from the same $\mathcal{Y}_{k-1}$; there is no fixed training set.

- (ii) *Adaptive step.* The next level t_k is fixed by the adaptive selection scheme (Algorithm 3): starting from an initial guess t_{init} and shrinking, it takes the largest step below the guess whose classical SMC ESS from μ_{k-1} to μ_k , run on the selection pool of step (i), stays above the floor τ . The first stage uses the safe start $t_{\text{init}} = t_{\text{safe}}$: the leading increment faces the largest deformation, and an over-aggressive start is exposed only after a full training, so the first stage begins from a level it is sure to handle. Every later stage extrapolates the accepted increments, $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$ with enlarge factor Γ : the step sizes are allowed to grow, since only the first few stages are essentially difficult. This sets the stage target $\mu_k \propto \exp(-U_k)$ with $U_k = (1 - t_k)U_0 + t_k U$.
- (iii) *Wide-coverage set.* A wide-coverage set $\hat{\mathcal{Y}}_k$ of P samples, distributed as $\hat{\mu}_k$, is drawn by quench and temper (Algorithm 2) on the stage target U_k , spreading samples across all of its modes. It supplies the wide-coverage half of the mixture term in the training loss.
- (iv) *Training loss.* The flow G_k is trained by the X-regularized forward KL of Section 3.1, specialized to the stage $\mu_{k-1} \rightarrow \mu_k$ at the balanced hyperparameters $\lambda = 1$, $\alpha = \beta = 1/2$. With model $\nu_k = (G_k^{-1})_{\#}\mu_{k-1}$ and its stop-gradient copy $\bar{\nu}_k$, the stage log-ratio is

$$z_k(y) = U_{k-1}(G_k(y)) - U_k(y) - \log |\det J_{G_k}(y)|,$$

and the loss reads

$$\mathcal{L}[G_k] = \underbrace{\mathbb{E}_{y \sim \mu_k} [z_k(y)]}_{\text{KL}(\mu_k \parallel \nu_k)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu_k} [z_k(y) - z_k(y')]}_{X_{\mu_k}(\mu_k \parallel \nu_k)} + \underbrace{\mathbb{E}_{y, y' \sim (\hat{\mu}_k + \bar{\nu}_k)/2} [z_k(y) - z_k(y')]}_{X_{(\hat{\mu}_k + \bar{\nu}_k)/2}(\mu_k \parallel \nu_k)}, \quad (28)$$

its μ_k -expectations estimated by annealed importance sampling: the level- $(k-1)$ batch $\mathcal{X}$ is pushed to $\mathcal{X}_{\nu} = G_k^{-1}(\mathcal{X})$, distributed as ν_k , and annealed to μ_k by Algorithm 1 with $G = G_k$, sharing the

ladder length M and rejuvenation parameters with the SMC of step (ii); the wide-coverage measure $\hat{\mu}_k$ is realized by the set $\tilde{\mathcal{Y}}_k$ of step (iii).

- (v) *Validation set update.* Once G_k is trained, the validation set is advanced from $\mathcal{Y}_{k-1}$ to $\mathcal{Y}_k \approx \mu_k$ by importance sampling. Each $y \in \mathcal{Y}_{k-1}$ is pushed through the trained inverse to $\tilde{y} = G_k^{-1}(y)$, a sample of $\nu_k = (G_k^{-1})_{\#}\mu_{k-1}$, and carries the importance weight

$$w(\tilde{y}) = \frac{\mu_k(\tilde{y})}{\nu_k(\tilde{y})} = \exp(z_k(\tilde{y})).$$

Its effective sample size $\text{ESS}(w)$ is the final, flow-based check on the increment: the adaptive selection of step (ii) sizes t_k from a classical SMC ESS that uses no QT samples and may therefore report a fake ESS, whereas $\text{ESS}(w)$ measures the trained flow directly. If $\text{ESS}(w) < \tau_v$ (we use $\tau_v = 0.3$), the increment was too aggressive: we discard G_k , shrink it to $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$, and repeat steps (iii)–(v) at the reduced step. Otherwise the validation set is advanced to $\mathcal{Y}_k = \text{langevin}(\text{resample}(\tilde{\mathcal{Y}}, w), U_k) \approx \mu_k$ (resample $\tilde{\mathcal{Y}}$ by its weights w , then rejuvenate with a short Langevin chain on U_k).

The trained ladder is a Boltzmann generator for μ : starting from $x \sim \mu_0$, each stage applies its map G_k^{-1} and then reweights and resamples by the importance weights of Section 3.4, exactly as in step (v), carrying the samples from the source to the target one bridge at a time.

5 Numerical experiments

5.1 2D benchmark distributions

We test the training scheme on six 2D targets, each chosen to isolate a different failure mode of the forward KL rather than to match the GMM/DW community suite (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024): Threewell (three asymmetric wells of a quartic-plus-sinusoid potential, with the narrow source far from all three), Himmelblau (four isolated modes), Annulus (a thin one-dimensional ring of minima together with an isolated central minimum), Python (a thin 1D-manifold version of the Python logo built from 1024 stroke Gaussians traced along the skeleton of the two interlocking snake bodies, plus two larger eye Gaussians), Chessboard (a Gaussian mixture on 26 of the 32 black squares of an 8×8 grid, with six cells removed to break the chessboard symmetry), and Sparse (a four-mode Gaussian mixture with a dense pair near the origin and two isolated modes in opposite corners, several source-standard-deviations away, that the narrow source tail cannot reach). The potentials are listed in Table 1.

For all six targets the source μ_0 is an isotropic Gaussian $\mathcal{N}(0, \sigma^2 I)$ with σ between 0.25 and 2.0; on Threewell it is deliberately narrow and centered at the origin, far from all three wells, so the initial pushforward covers none of them. The flow is a neural spline flow with 6 coupling transforms, 32 bins, and hidden widths (128, 128), on a square box matched to each target. Training uses Adam (learning rate 10^{-3} to 5×10^{-3}) for 1000 steps (2000 on Python, whose narrow tubular support takes longer to saturate), with batch size 500, enlarged to 1000–2000 on the multi-mode Threewell, Chessboard, and Sparse so a single batch can cover all modes. At each step the AIS surrogate is one importance-weighted resampling pass against the current pushforward followed by 50 Langevin steps. The loss coefficients are the preferred default of Section 3.1, $\lambda = 1$ and $\alpha = \beta = 1/2$, and the wide-coverage measure $\hat{\mu}$ is built once before training by QT (Algorithm 2) and recycled with one Langevin rejuvenation per step.

For each target we compare four losses obtained by progressively adding one X functional at a time: the bare forward KL (2); $\text{KL} + X_\mu$ with $\lambda = 1$; $\text{KL} + X_\mu + X_{\hat{\mu}}$ with $\alpha = 1$, $\beta = 0$; and the preferred mixture $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ with $\alpha = \beta = 1/2$.

Threewell. The quartic-plus-sinusoid potential has three asymmetric wells. Forward KL and $\text{KL} + X_\mu$ lock onto only two of them and never discover the third, yet their ESS stays close to one: a fake ESS instance, the flow omitting an entire mode while reporting near-uniform weights on the support it reaches. Adding $X_{\hat{\mu}}$ pulls mass onto the QT-discovered third well, and the mixture attains the same full coverage at the highest full-coverage ESS with the cleanest wells of the four (Figure 1a).

Algorithm 4: Adaptive-temperature Boltzmann generator training

Input: potentials U_0, U , pool size P , batch size B , validation ESS floor $\tau_v = 0.3$, shrink factor $\gamma = 0.7$,
enlarge factor $\Gamma = 2$, safe start $t_{\text{safe}} = 0.25$

Output: ladder $0 = t_0 < \dots < t_K = 1$ and flow maps $\{G_k\}_{k=1}^K$ with $(G_k^{-1})_{\#} \mu_{k-1} \approx \mu_k$

// validation set, $\mathcal{Y}_k \approx \mu_k$

```
1  $\mathcal{Y}_0 \leftarrow$  samples from  $\mu_0$ ,  $t_0 \leftarrow 0$ ,  $k \leftarrow 0$ ;  
2 while  $t_k < 1$  do  
3    $k \leftarrow k + 1$ ;  
4   Form the stage source potential,  $U_{k-1} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U$ ;  
   // (i) source samples (selection pool)  
5   Draw a pool  $\mathcal{P}$  of  $P$  samples from  $\mathcal{Y}_{k-1}$  and rejuvenate,  $\mathcal{P} \leftarrow \text{langevin}(\mathcal{P}, U_{k-1})$ ;  
   // (ii) adaptive temperature selection  
6   if  $k = 1$  then  
7     Set  $t_{\text{init}} = t_{\text{safe}}$ ;  
8   else  
9     Set  $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$ ;  
10  end  
11  Choose  $t_k$  by adaptive selection (Algorithm 3) on the pool  $\mathcal{P}$  with initial guess  $t_{\text{init}}$ ;  
12  Form the target potential  $U_k \leftarrow (1 - t_k)U_0 + t_kU$ ;  
13  repeat  
14    Initialize  $G_k$  to the identity map;  
    // (iii) wide-coverage set  
15    Draw the wide-coverage set of  $P$  samples by quench and temper,  $\hat{\mathcal{Y}}_k \leftarrow \text{QT}(U_k)$  (Algorithm 2);  
    // (iv) X-regularized training loss  
16    for each gradient step do  
17      Draw a training batch of  $B$  samples  $\mathcal{X}$  from the validation set  $\mathcal{Y}_{k-1}$ ;  
18      Form the detached model batch,  $\tilde{\mathcal{X}}_\nu \leftarrow G_k^{-1}(\mathcal{X})$ , distributed as  $\nu_k = (G_k^{-1})_{\#} \mu_{k-1}$ ;  
19      Anneal from  $\tilde{\mathcal{X}}_\nu$  to  $\mathcal{X}_\mu$  via AIS (Algorithm 1, from  $\nu_k$  to  $\mu_k$ );  
20      Draw a wide-coverage batch  $\hat{\mathcal{X}}_\mu \subset \hat{\mathcal{Y}}_k$  and rejuvenate,  $\hat{\mathcal{X}}_\mu \leftarrow \text{langevin}(\hat{\mathcal{X}}_\mu, U_k)$ ;  
21      Update  $G_k$  by a gradient step on the X-regularized forward KL (28), from  $\mathcal{X}_\mu$  and the balanced  
      mixture of  $\hat{\mathcal{X}}_\mu$  and  $\tilde{\mathcal{X}}_\nu$ ;  
22    end  
    // (v) validation set update  
23    Push the validation set  $\tilde{\mathcal{Y}} \leftarrow G_k^{-1}(\mathcal{Y}_{k-1})$ ;  
24    Compute the weights  $w \leftarrow \exp(U_{k-1}(\mathcal{Y}_{k-1}) - U_k(\tilde{\mathcal{Y}}) - \log |\det J_{G_k}(\tilde{\mathcal{Y}})|)$ ;  
25    if  $\text{ESS}(w) < \tau_v$  then  
26      // abort the step and shrink toward  $t_{k-1}$   
       $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$ , then re-form  $U_k \leftarrow (1 - t_k)U_0 + t_kU$ ;  
27    else  
28      Resample  $\mathcal{Y}_k \leftarrow \text{resample}(\tilde{\mathcal{Y}}, w)$ ;  
29      Employ Langevin rejuvenation to obtain  $\mathcal{Y}_k \leftarrow \text{langevin}(\mathcal{Y}_k, U_k)$ ;  
30    end  
31  until  $\text{ESS}(w) \geq \tau_v$ ;  
32 end  
33  $K \leftarrow k$ ;  
34 return  $\{t_k\}_{k=0}^K$  and  $\{G_k\}_{k=1}^K$ ;
```

name	potential $U(x_1, x_2)$	minima
Threewell	$6[(x_1^2 - 1)^2 + (x_2^2 - 1)^2 + \sin(x_1 + 2x_2)]$	three asymmetric wells
Himmelblau	$(x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$	four isolated points
Annulus	$20(r^6 - 8r^4 + 16r^2 + 1)^{\frac{1}{3}}, r^2 = x_1^2 + x_2^2$	$r = 0$ and ring at $r = 2$
Python	Python-like Gaussian mixture	1024 stroke centers + 2 eye centers
Chessboard	Chessboard-like Gaussian mixture	26 centers
Sparse	Sparse Gaussian mixture	2 dense + 2 isolated centers

Table 1: Six 2D benchmark targets.

Himmelblau. Forward KL and $\text{KL} + X_\mu$ both cover only two of the four modes, yet the ESS gives no warning: $\text{KL} + X_\mu$ posts the highest nominal ESS of the four while missing half the modes, a fake ESS instance. Adding $X_{\hat{\mu}}$ or the mixture recovers full coverage and ties at the best full-coverage ESS, the mixture pushforward being the visibly cleaner of the two (Figure 1b).

Annulus. All four methods reach full coverage, so the discriminator is the flow shape and ESS. Forward KL alone leaves visible axis-aligned filaments across the central barrier; adding X_μ tightens the pushforward to a clean ring-plus-point geometry and sharply lifts the ESS, and the mixture attains the best ESS of the four (Figure 2a). This isolates the role of X_μ as a shape regularizer, penalizing pointwise log-ratio variation that the global forward KL leaves untouched.

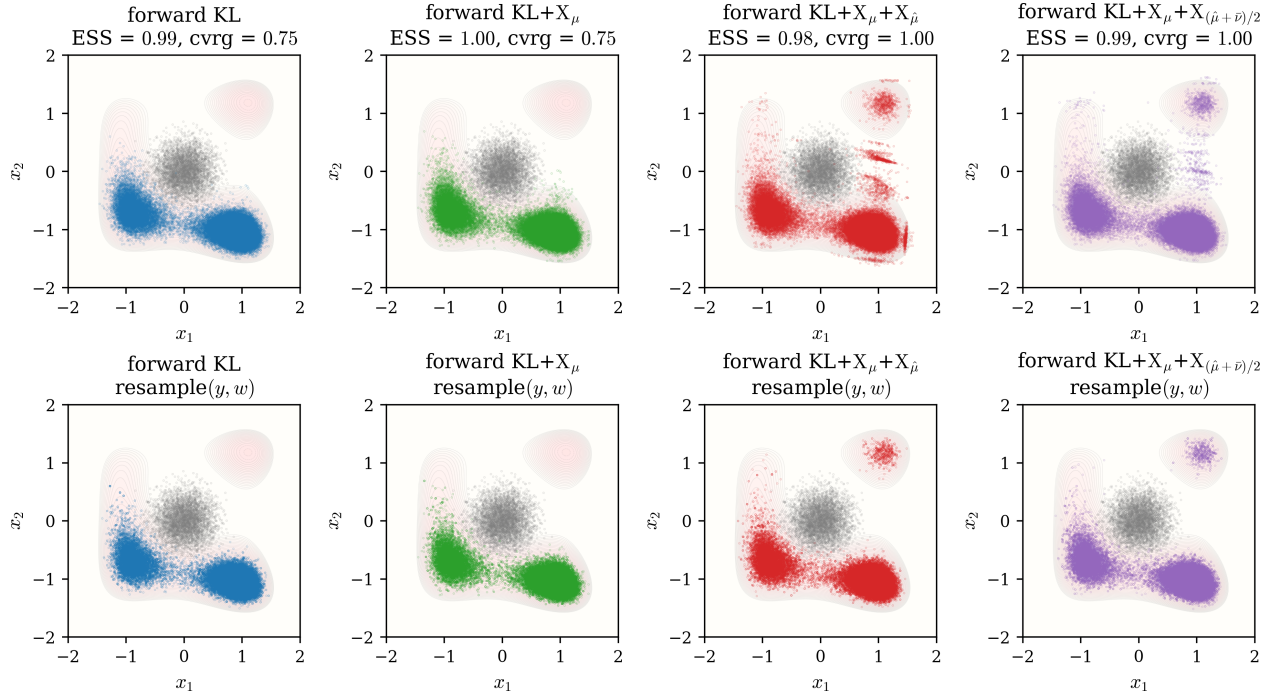
Python. The Python target lives on an essentially one-dimensional manifold, the skeleton of the two interlocking snake bodies of the Python logo. All four methods reach full coverage and the ESS plateaus well below 1. The mode-discovery role of $X_{\hat{\mu}}$ is inactive here, yet the X-regularized losses still improve on bare forward KL, and the mixture reaches the best ESS of the four, leading the next-best X variant by ≈ 0.10 (Figure 2b) — consistent with $\bar{\nu}$ anchoring the off-manifold profile of ν .

Chessboard. Coverage is full for all four methods, but mode structure and ESS vary. Forward KL leaks mass into the six removed cells, and neither X_μ nor $X_{\hat{\mu}}$ stops the leakage ($\hat{\mu}$ alone is sample-blind to the removed cells). The mixture resolves it: the $\bar{\nu}$ half sees the holes whenever ν leaks mass there, and the cross contributions $|\log(\mu/\nu)(y) - \log(\mu/\nu)(y')|$ with $y \sim \hat{\mu}$ and $y' \sim \bar{\nu}$ anchor in-cell and out-of-cell log-ratios to each other, placing clean mass on the filled cells at the best ESS by a clear margin (Figure 3).

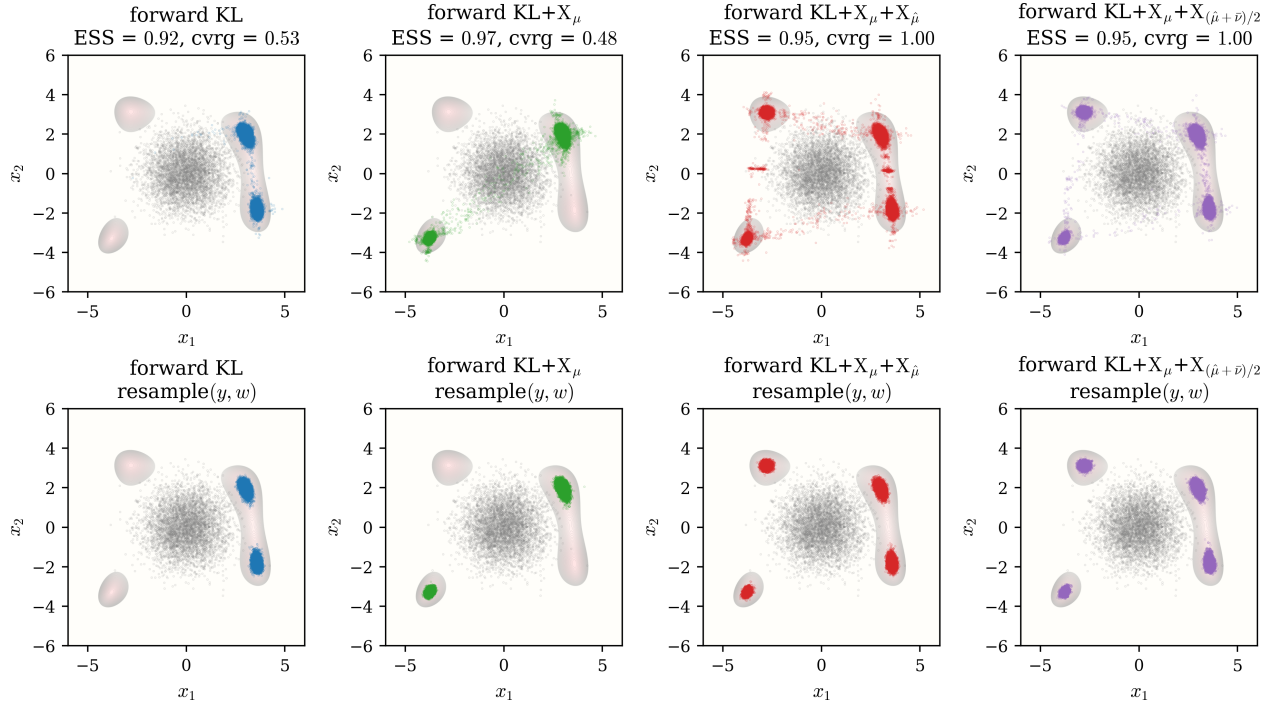
Sparse. A four-mode Gaussian mixture with a dense central pair and two isolated corner modes the narrow source cannot reach. Forward KL and $\text{KL} + X_\mu$ lock onto the central pair and miss the corners at high, stable ESS, another fake ESS instance. Adding $X_{\hat{\mu}}$ restores full coverage but, lacking the $\bar{\nu}$ half, overfits $\hat{\mu}$ and leaves artificial bridges of mass between the modes; the mixture also reaches full coverage while removing these bridges, at the best full-coverage ESS (Figure 4).

Across the six targets two regimes emerge. On the mode-discovery targets (Threewell, Himmelblau, Sparse) the ESS stays high for every loss and is therefore useless on its own: forward KL and $\text{KL} + X_\mu$ post ESS up to 1.00 while silently missing whole modes, and only the $X_{\hat{\mu}}$ and mixture variants restore full coverage. On the shape, manifold, and leakage targets (Annulus, Python, Chessboard) coverage is already complete and ESS becomes the discriminator: X_μ helps on Annulus and Python but is flat on Chessboard, whereas the mixture improves on bare forward KL by +0.31, +0.22, and +0.07 and attains the best ESS on all three. Overall the mixture is the only loss among the four compared that reaches full mode coverage and the best (or tied-best) full-coverage ESS on every target, and its ESS curves are visibly smoother than bare forward KL.

The mixture loss $\mathcal{L} = \text{KL}(\mu\|\nu) + X_\mu(\mu\|\nu) + X_{(\hat{\mu}+\bar{\nu})/2}(\mu\|\nu)$ with $\lambda = 1$ and $\alpha = \beta = 1/2$ is the recommended default. It absorbs the mode-discovery role of $X_{\hat{\mu}}$ while suppressing the off-support leakage that $X_{\hat{\mu}}$ alone can introduce when the target has holes interleaved with its modes.

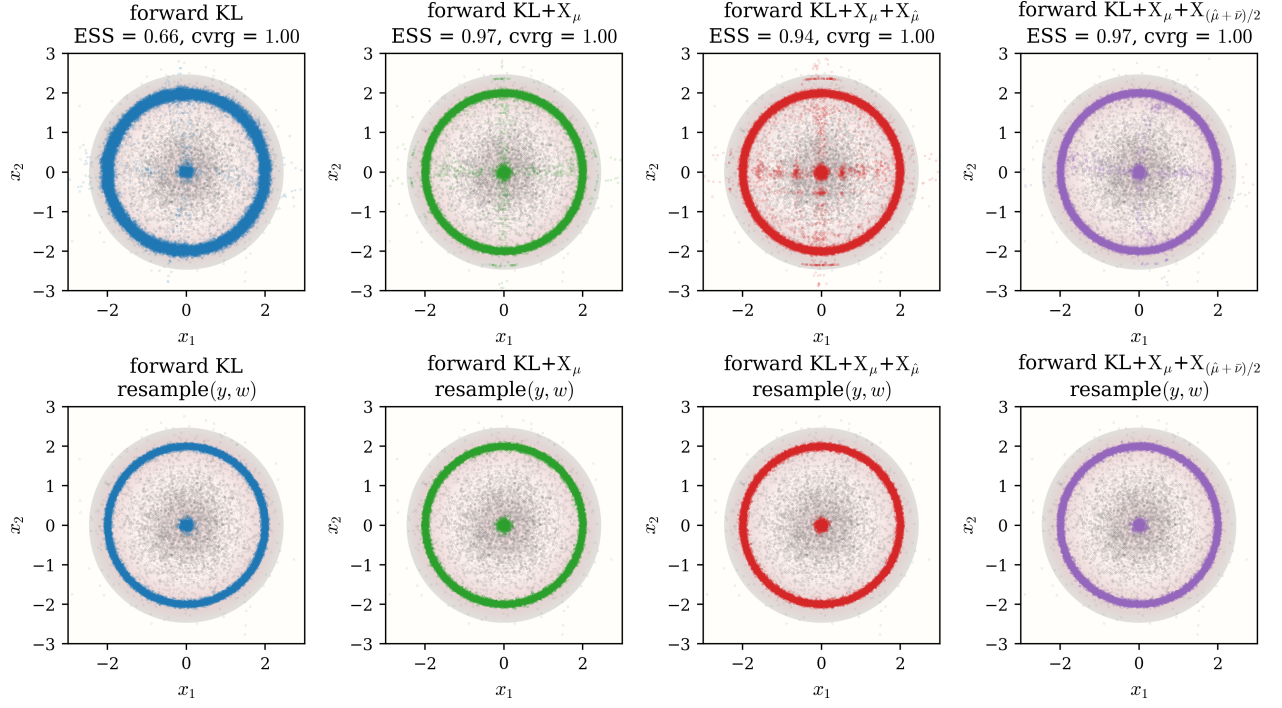


(a) Threewell target.

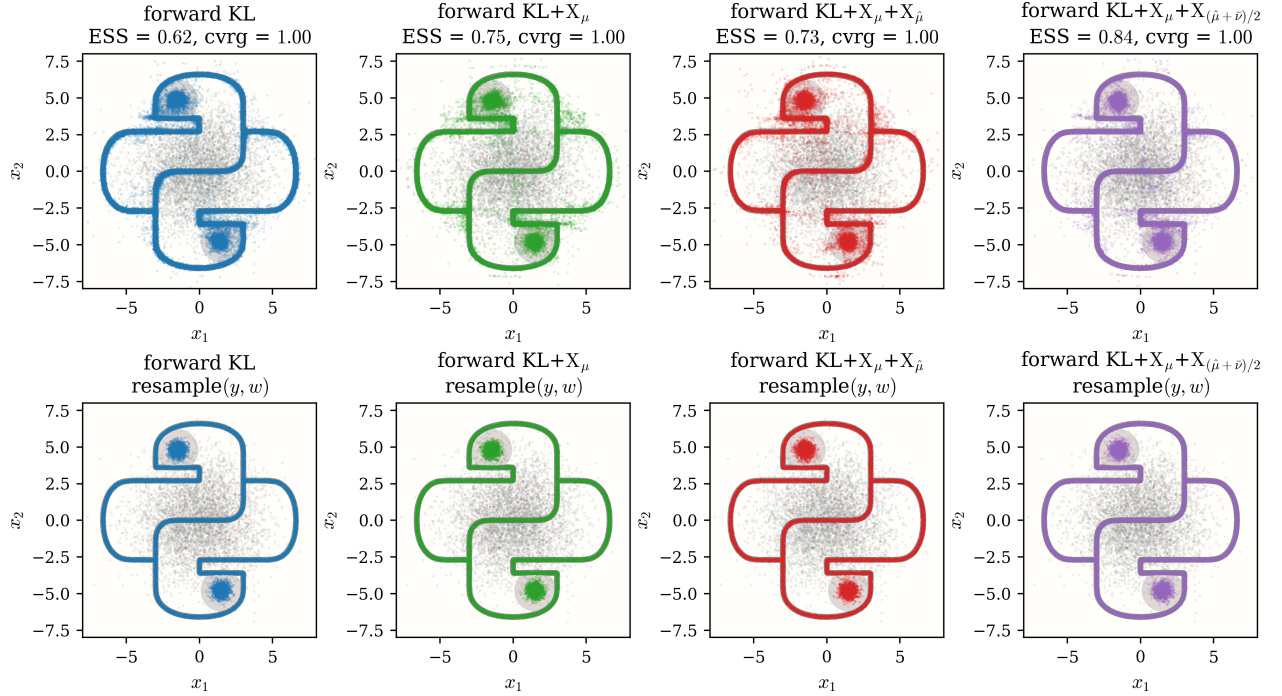


(b) Himmelblau target.

Figure 1: Mode-discovery targets. In each panel, Row 1 is the pushforward $y = G^{-1}(x)$ and Row 2 is $\text{resample}(y, w)$ with Langevin rejuvenation. Columns left to right: forward KL, $\text{KL}+X_{\mu}$, $\text{KL}+X_{\mu}+X_{\hat{\mu}}$, $\text{KL}+X_{\mu}+X_{(\hat{\mu}+\bar{\nu})/2}$. Each pushforward title reports the final ESS and the $\kappa = 5$ coverage against $\hat{\mu}$. Forward KL and $\text{KL}+X_{\mu}$ miss modes at near-perfect ESS, while $X_{\hat{\mu}}$ and the mixture recover full coverage.



(a) Annulus target.



(b) Python target.

Figure 2: Shape and manifold targets. Rows and columns as in Figure 1.

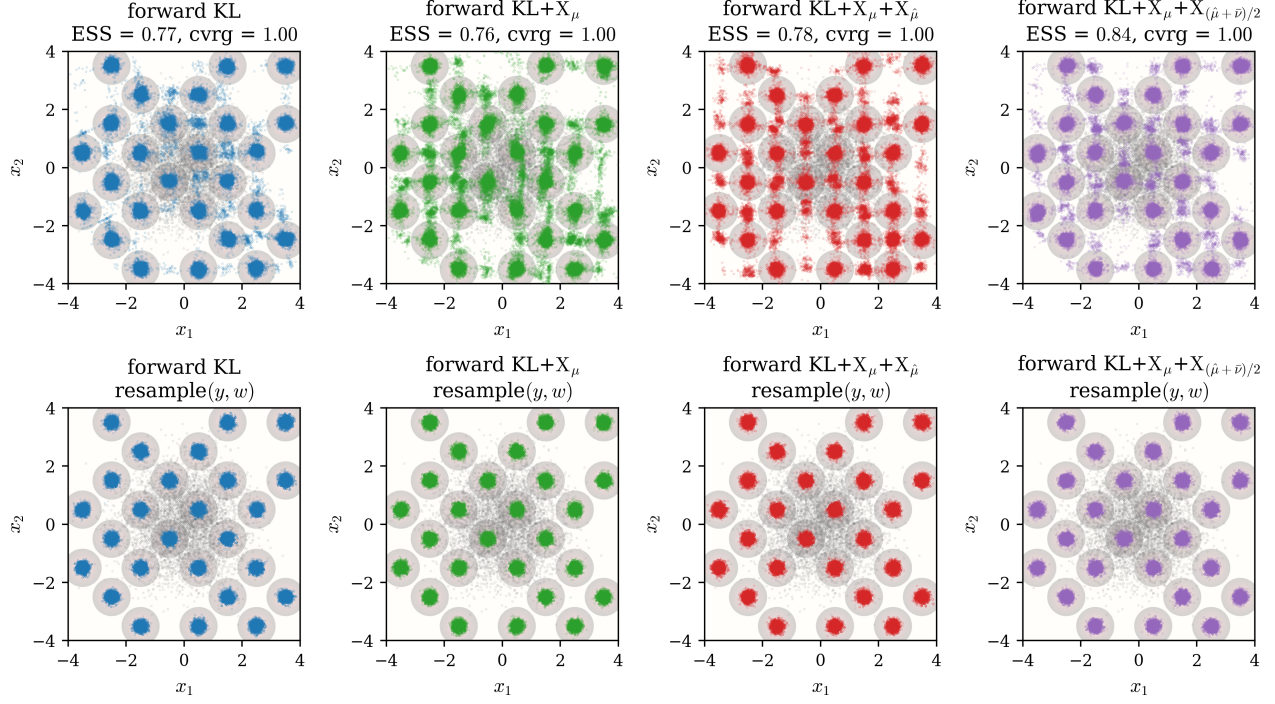


Figure 3: Chessboard target. Row 1: pushforward $y = G^{-1}(x)$. Row 2: $\text{resample}(y, w)$ with Langevin rejuvenation. Columns as in Figure 1.

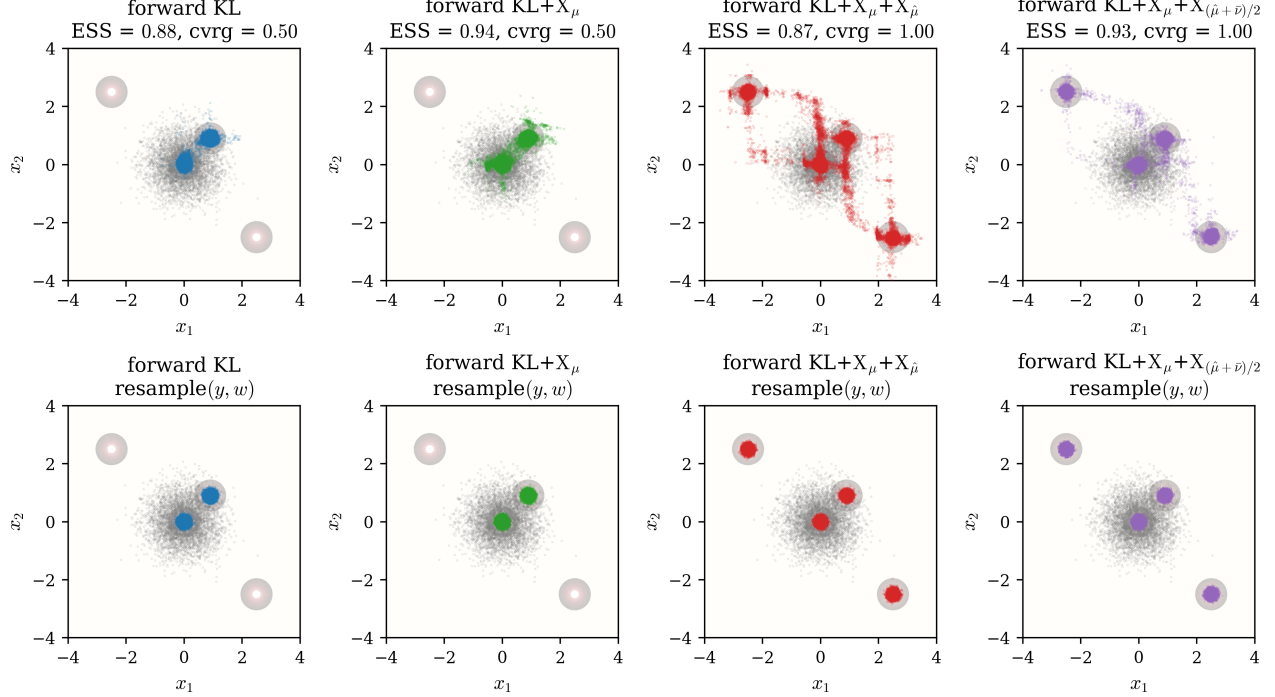


Figure 4: Sparse target. Row 1: pushforward $y = G^{-1}(x)$. Row 2: $\text{resample}(y, w)$ with Langevin rejuvenation. Columns as in Figure 1. Forward KL and $\text{KL}+X_{\mu}$ collapse onto the two central modes and miss both isolated corners at high, stable ESS (a fake ESS instance); $X_{\hat{\mu}}$ recovers all four modes but leaves artificial bridges of mass between them, while the mixture recovers them cleanly.

target	KL	KL+X _μ	KL+X _μ +X _{μ̂}	KL+X _μ +X _{(μ̂+ν̄)/2}
Threewell	0.99/0.75	1.00/0.75	0.98/ 1.00	0.99/1.00
Himmelblau	0.92/0.53	0.97/0.48	0.95/1.00	0.95/1.00
Annulus	0.66/1.00	0.97/1.00	0.94/1.00	0.97/1.00
Python	0.62/1.00	0.75/1.00	0.73/1.00	0.84/1.00
Chessboard	0.77/1.00	0.76/1.00	0.78/1.00	0.84/1.00
Sparse	0.88/0.50	0.94/0.50	0.87/ 1.00	0.93/1.00

Table 2: Final ESS / $\kappa = 5$ coverage against $\hat{\mu}$ for each loss on each 2D target (best full-coverage ESS per row in bold). On the mode-discovery targets (Threewell, Himmelblau, Sparse) forward KL and X_μ post high ESS while silently missing modes, and only X_{μ̂} and the mixture restore full coverage. On the remaining targets coverage is complete and X_μ improves the ESS except on Chessboard, where only the mixture does. The mixture is the only variant that attains the best (or tied-best) full-coverage ESS on every target while always retaining full coverage.

5.2 Bayesian inverse problem: sensor-array source localization

Our next benchmark is a physical Bayesian inverse problem where the multimodality comes from a symmetry of the measurement, not from a hand-built mixture. The setting is source localization, a standard problem in acoustics and signal processing: a line of fixed sensors records the combined signal of several identical point sources, and we infer the source positions from the noisy readings. The unknown is the vector of source positions $\theta = (\theta_1, \dots, \theta_n) \in \mathbb{R}^n$ for $n = 3$ sources. A source at θ_i leaves a Gaussian bump of width $\ell = 0.8$ at each sensor and the sensors read the sum, so with $J = 25$ sensors at fixed positions $s_1, \dots, s_J$ equispaced on $[-6, 6]$ the reading at sensor j is

$$g_j(\theta) = \sum_{i=1}^n \exp\left(-\frac{(s_j - \theta_i)^2}{2\ell^2}\right).$$

The data are these readings at the true positions $\theta^* = (5, 0, -5)$ with i.i.d. Gaussian noise,

$$y_j = g_j(\theta^*) + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma_{\text{obs}}^2), \quad \sigma_{\text{obs}} = 0.3,$$

fixed across all losses. The prior on the positions is a centered isotropic Gaussian $\mu_0 = \mathcal{N}(0, \sigma_{\text{prior}}^2 I_n)$ with $\sigma_{\text{prior}} = 1.6$, which is also the flow’s source, and the negative log-posterior is the prior plus the data fit,

$$U(\theta) = \frac{1}{2\sigma_{\text{obs}}^2} \sum_{j=1}^J (g_j(\theta) - y_j)^2 + \frac{1}{2\sigma_{\text{prior}}^2} \|\theta\|^2. \quad (29)$$

Because the sources are identical, g_j is a sum over them and is unchanged by any permutation of the coordinates, $g_j(\theta_{\pi(1)}, \dots, \theta_{\pi(n)}) = g_j(\theta)$ for $\pi \in S_3$; the prior is permutation-invariant too, so $U(\pi \cdot \theta) = U(\theta)$. The posterior therefore has exactly $3! = 6$ equal, well-separated modes (the permutations of θ^*), and the centered prior sits on their common symmetry point at the origin, about 4.4 prior standard deviations from every mode ($\|\theta^*\| = 7.07$). The prior tail reaches none of them, so a flow trained by bare forward KL only sees the one or two modes its pushforward first lands on and collapses there. We use the same flow, training, and diagnostics as Section 5.1.

Table 3 and Figure 5 report the comparison. Bare forward KL covers only 2 of the 6 modes at ESS 0.90, and adding X_μ posts the highest ESS of the four, 0.98, while collapsing onto a single mode — the ESS improves exactly as coverage gets worse. The QT-driven X_{μ̂} and the mixture both recover all 6 modes, with mass split evenly, at ESS 0.91 and 0.93.

This benchmark is intentionally low-dimensional, and the numbers come from a single training seed; which modes a collapsed method happens to keep is set by the random path, not by the loss.

loss	final ESS	modes found
forward KL	0.904	2
KL+ X_μ	0.979	1
KL+ X_μ + $X_{\hat{\mu}}$	0.913	6
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.934	6

Table 3: Sensor-array source-localization benchmark (29) with $n = 3$ sources and $3! = 6$ permutation posterior modes. Bare forward KL and KL+ X_μ post a fake ESS (the highest, 0.98, for X_μ) while covering only 1–2 of the six equal-mass modes; the QT-driven $X_{\hat{\mu}}$ and mixture variants restore full 6-mode coverage. Modes found is the number of the six permutation modes that hold at least 5% of the pushforward mass.

5.3 High-dimensional product multi-well

To test the framework beyond two dimensions we construct a family of product multi-well targets in dimension $d = 2^k$, playing here the role that double-well systems play in the diffusion-sampler literature (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024), but with an analytically known mode structure and a diagonal optimal transport map so that training failures are unambiguously training failures and not capacity failures of the flow,

$$U(x) = \frac{1}{2} \sum_{i=1}^{2^k} x_i^2 + 12 \sum_{i=1}^k \exp(-x_i^2), \quad (30)$$

a double well in each of the first k coordinates (minima at $\pm\sqrt{\ln 24} \approx \pm 1.78$, separated by a barrier of height ≈ 9.9) and a standard Gaussian in the remaining $2^k - k$. The source is the standard Gaussian $\mu_0 = \mathcal{N}(0, I_d)$, whose potential is the quadratic part of U . Because U is separable, the target factorizes into k identical double wells and $2^k - k$ Gaussians, so it has exactly $2^k = d$ modes (the number of modes equals the dimension), and the optimal transport map is diagonal. The source sits exactly on the barrier of every double well, so the untrained pushforward covers none of the 2^k mode combinations. We sweep $k = 1, \dots, 8$, i.e. $d = 2, 4, 8, 16, 32, 64, 128, 256$.

The flow is a neural spline flow on $[-4, 4]^d$ with 6 coupling transforms, 16 bins, and hidden widths (256, 256). Training uses Adam at learning rate 10^{-3} with batch size 200 (forward KL is highly sensitive to the batch size, so a small batch sharpens the contrast between the losses), drawn from a source set of $10^4 \cdot 2^k$ samples; the number of steps decreases with d from 3000 ($d \leq 4$) to 1200 ($d \geq 64$). The μ -surrogate is the same one-step importance-resampling pass followed by 50 Langevin steps used in the 2D experiments, and $\hat{\mu}$ is built once by QT from $100 \cdot 2^k$ samples. The four losses are exactly those of Section 5.1, with $\lambda = 1$ and $\alpha = \beta = 1/2$.

Table 4 reports the final ESS of each loss across the eight dimensions. Unlike the 2D mode-discovery targets, here the flow reaches all 2^k modes for every loss (each mode is filled to within a few percent of its uniform 2^{-k} share, and the $\kappa = 5$ coverage against $\hat{\mu}$ stays ≥ 0.94 throughout), so this is the calibration regime of Annulus, Python, and Chessboard rather than the fake ESS regime, and ESS is the sole discriminator. Three trends are clear. First, the bare forward KL degrades monotonically with dimension, from 0.96 at $d = 2$ to 0.38 at $d = 256$, as the log-weight variance accumulates one double well at a time. Second, the X functionals counteract this and the gap widens with dimension: negligible at $d = 2$, it reaches +0.21–0.22 at $d = 64$ and 128, and +0.18 at $d = 256$ where the wide-coverage $X_{\hat{\mu}}$ becomes the best loss. Third, the QT-driven losses ($X_{\hat{\mu}}$ and the mixture) overtake X_μ alone at high dimension: X_μ tracks the mixture closely through $d = 64$ but falls back to the bare KL level by $d = 256$ (0.33 vs 0.38), because it reweights the live one-step IS surrogate whose pushforward is poorly formed early in training, while $X_{\hat{\mu}}$ and the mixture hold their lead because the fixed QT set keeps the flow anchored on the wells even as the log-weight variance grows.

At $d = 256$ ($k = 8$), although the nominal dimension is 256 the *essential* dimension of the target is only $k = 8$: the potential (30) is a double well in the first 8 coordinates and a standard Gaussian in the remaining 248, so all the multimodality lives in an 8-dimensional subspace (while the log-weight variance, and hence the ESS, still loads on all 256). Here we train a single flow and sweep the AIS-surrogate ladder length $M \in \{1, 2, 4, 8\}$ that builds the μ -surrogate for the mixture loss (this is the inner ladder of Algorithm 1,

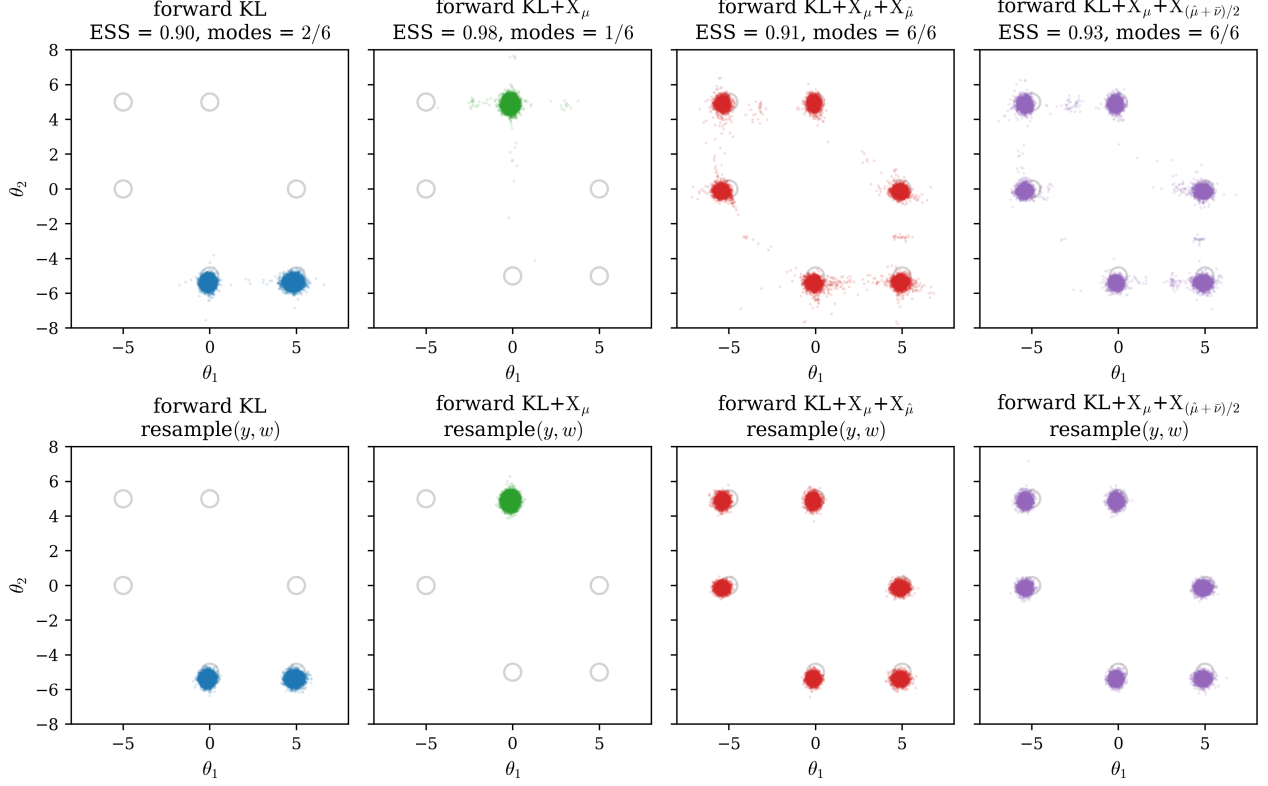


Figure 5: Sensor-array posterior, pushforward positions projected onto (θ_1, θ_2) . Row 1: raw pushforward $y = G^{-1}(x)$; Row 2: $\text{resample}(y, w)$ with Langevin rejuvenation. Columns left to right: forward KL, $\text{KL}+X_\mu$, $\text{KL}+X_\mu+X_{\hat{\mu}}$, $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$. Light-gray circles mark the six permutation mode centers (the permutations of θ^*); each title reports the final ESS and the modes found out of six. Forward KL and $\text{KL}+X_\mu$ collapse onto 1–2 modes at high ESS, while $X_{\hat{\mu}}$ and the mixture fill all six.

distinct from the temperature ladder of Section 4). Increasing M accelerates convergence in the first training stage (a longer ladder leaves the low-ESS plateau sooner), but does not affect the long-time limit: the ESS curves differ only in the first stage and settle at a comparable value (Figure 6). The ladder length is therefore a first-stage accelerator rather than a determinant of the final flow quality.

5.4 Boltzmann generator on p -state clock model

The preceding experiments train a single flow map; this experiment exercises the full adaptive-temperature procedure of Algorithm 4, contrasting the X -regularized stage loss (28) against the bare forward KL through an otherwise identical pipeline. The target is the p -state clock model on a periodic 8×8 square lattice, one angle per site,

$$U(\theta) = -J \sum_{\langle j,l \rangle} \cos(\theta_j - \theta_l) - h \sum_{j=1}^D \cos(p\theta_j), \quad \theta \in [-\pi, \pi)^D, \quad D = 64, \quad (31)$$

with nearest-neighbor coupling $J = 1$, $\mathbb{Z}_p$ anisotropy $h = 1/2$, and $p = 6$ clock states; for $p \geq 5$ the model hosts two Berezinskii–Kosterlitz–Thouless transitions enclosing a critical quasi-long-range-ordered phase (José *et al.*, 1977). The cold measure $\propto e^{-U}$ splits into $p = 6$ equivalent sectors labeled by the phase of the global magnetization

$$m(\theta) = \frac{1}{D} \sum_{j=1}^D e^{i\theta_j}, \quad s(\theta) = \left\lfloor \frac{p \arg m(\theta)}{2\pi} \right\rfloor \bmod p \in \{0, \dots, 5\},$$

loss \ d	2	4	8	16	32	64	128	256
forward KL	0.958	0.937	0.935	0.871	0.790	0.669	0.547	0.384
KL+ X_μ	0.998	0.983	0.972	0.961	0.893	0.845	0.657	0.333
KL+ X_μ + $X_{\hat{\mu}}$	0.987	0.990	0.973	0.955	0.948	0.835	0.750	0.565
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.986	0.992	0.975	0.969	0.953	0.878	0.767	0.543

Table 4: Final ESS of each loss on the product multi-well target (30) across dimension $d = 2^k$, $k = 1, \dots, 8$ (best per column in bold). All four losses reach full mode coverage at every d , so ESS is the discriminator. Forward KL degrades steadily with d ; the QT-driven losses ($X_{\hat{\mu}}$ and the mixture) are best or tied-best for every $d \geq 4$. The mixture leads from $d = 4$ to $d = 128$, with a margin of +0.21–0.22 over bare forward KL at $d = 64, 128$; at $d = 256$ the wide-coverage $X_{\hat{\mu}}$ alone takes the lead by +0.18 over bare KL, while X_μ alone falls back to the bare KL level.

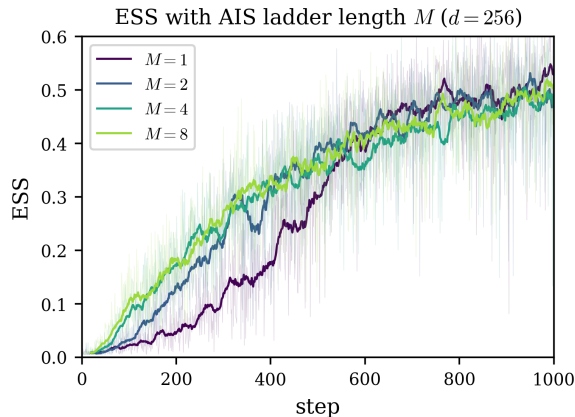


Figure 6: ESS trajectory across training steps for the mixture loss $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$ on the product multi-well at $d = 256$, as the AIS ladder length M that builds the μ -surrogate is varied over $\{1, 2, 4, 8\}$. A longer ladder M accelerates the first-stage convergence (the $M = 1$ one-step surrogate is slowest to leave the low-ESS plateau), but does not change the long-time limit: the curves differ only early and settle at a comparable ESS. Faint lines are the raw per-step ESS; bold lines are a moving average.

where $\lfloor \cdot \rfloor$ denotes rounding to the nearest integer. The annealing path from the uniform measure to μ traverses the BKT window, where the correlation length grows rapidly and the six sectors deepen simultaneously. Figure 7 displays this structure on the trained validation set.

The source is the uniform measure on the torus, $U_0 = 0$, so $U_t = tU$; each stage map G_k is a circular neural spline flow on $[-\pi, \pi)^{64}$. The protocol is deliberately large-sample: a validation set of 10^6 samples, pool size $P = 2 \times 10^5$, safe start $t_{\text{safe}} = 0.2$, shared ladder length $M = 4$, and SMC floor $\tau = 0.7$, which trims over-aggressive guesses before any training is spent. We report three batch sizes $B = 10^3, 10^4, 10^5$, with 2000, 500, 200 Adam steps per stage; the remaining constants are those of Algorithm 4. Training carries a stability guard: a non-finite loss skips the gradient step under gradient clipping and a fresh batch is drawn, with a stage abort only after repeated skips. The two losses compared are the full stage loss (28) at the balanced hyperparameters $\lambda = 1$, $\alpha = \beta = 1/2$, and the bare forward KL, its first term alone; everything else is shared.

Table 5 reports the core metric, the validation ESS of step (v) on every accepted rung. Both losses complete the ladder at every batch size, along *identical* ladders with *identical* rejected rungs: the adaptive machinery converts loss quality into schedule, not into success or failure. Figure 8 shows the per-step training ESS on the first three stages of each run. The discrimination lives in the per-rung quality. On accepted rungs the X-regularized loss posts the higher validation ESS on stages 1–3 at all three batch sizes, +0.21/ +0.13/ +0.10 at $B = 10^3$, +0.17/ +0.11/ +0.03 at $B = 10^4$, +0.13/ +0.09/ +0.04 at $B = 10^5$; the gap closes to zero in the transition window, slightly reversing on a few late rungs, at equal wall time per pair. A generalization check pushes 10^6 independent source samples through the trained $B = 10^4$ ladders,

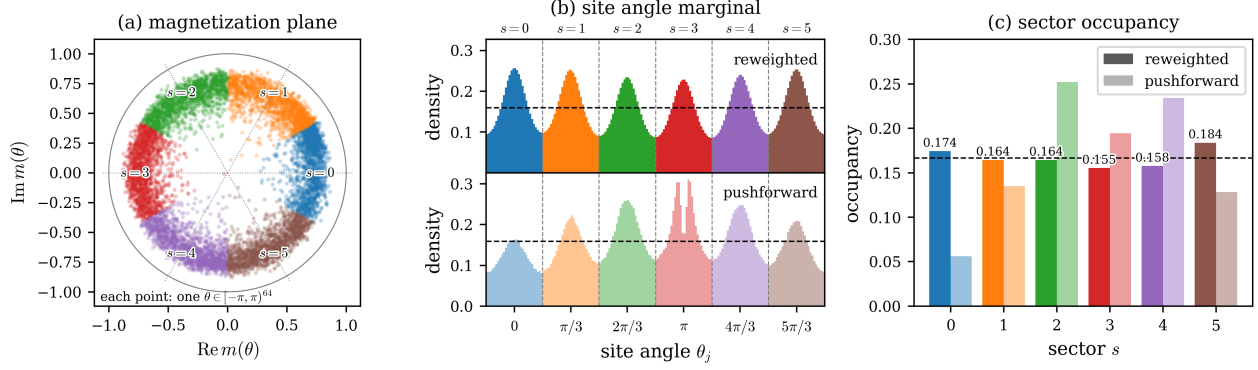


Figure 7: Structure of the cold clock measure (31) at $D = 64$, displayed on 10^6 reweighted samples: fresh source draws advanced through the trained ladder by the map–reweight–resample step (v), X-regularized loss at $B = 10^4$. (a) Global magnetization $m(\theta)$, colored by sector index $s(\theta)$, 1500 points per sector: the measure splits into $p = 6$ equivalent sectors with mean $|m| = 0.73$, well developed yet contiguous through the disordered center, so every training batch touches all sectors and the multimodality is soft. (b) Site-angle marginal: the anisotropy locks each site to the six clock angles $2\pi k/6$; dashed lines bound the six wells, colored as in (c), and the black line marks the uniform density $1/(2\pi)$; the pushforward marginal, the maps alone without reweighting, tracks the reweighted one closely. (c) Sector occupancy of reweighted and pushforward samples, the black line marking the uniform occupancy $1/6$: all six sectors are populated, and reweighting corrects the occupancy bias from total variation 0.18 to 0.025, uniform to ± 0.02 per sector.

stage by stage with the resampling of step (v): the fresh-sample per-stage ESS tracks the gate values within 0.02–0.10 for both losses, and the final sector occupancy is uniform to ± 0.02 per sector, total variation 0.025, so the gates measure generalization rather than memorization of the carried set.

The pattern is consistent with the theory: the X terms lower the asymptotic accuracy floor of each stage (Section 2.3), a larger training *limit*, but promise no faster transient, and the rejected-rung values confirm that the bare forward KL trains no slower. The anatomy of the ladder shows where the gain lives. The early bridges, $t \leq 0.66$, are far but smooth: a large, structurally simple deformation with a clean AIS signal, so the training loss is the binding constraint and the X terms win by up to +0.21. In the transition window, $t \approx 0.66$ –0.95, the selector halves the accepted increment, an empirical readout of the annealing bottleneck at the BKT crossing; there the ceiling is set by the critically slow Langevin rejuvenation and the shared network capacity, constraints common to both losses, so the gap collapses to zero. The final hop to $t = 1$ is the easiest rung for both. On this target the six sectors are never isolated (Figure 7(a)), so the mode-discovery channel of $X_{\hat{\mu}}$ is not load-bearing; the qualitative separations of Sections 5.1–5.3, where modes are isolated by construction, do not transfer to soft multimodality. What transfers is a uniform per-rung quality improvement at zero extra cost: the X-regularized loss was never worse than the bare forward KL where it mattered and never failed where the bare loss passed.

The trained ladder is used as a staged sampler: each stage applies its map G_k^{-1} and then reweights and resamples as in step (v), and Figure 7(c) shows the resulting occupancy, uniform to ± 0.02 . The residual is nonetheless well above the iid multinomial floor of 3×10^{-4} at 10^6 samples, and the carried training set itself is less uniform, at total variation 0.11: a side effect of finite-size sampling, not a training failure, and one that reweighting cannot simply repair. The weights of step (v) are pointwise density ratios, exchangeable within a sector and hence sector-blind: they neither detect the imbalance, seeded by the slight flow bias that the iid pushforward exposes, nor correct it; the cold Langevin rejuvenation transports no mass between sectors; and the final validation set is resampled, carrying no weights at all. The occupancy is thus unbiased in expectation over runs but unrestored within any single one, and the per-stage fluctuations accumulate along the ladder: in the fresh rebuild, essentially the whole residual arises at the two coldest rungs, where the ESS is lowest and resampling concentrates the lineages. Coverage, ESS, and occupancy evenness are three distinct axes of sample quality; the failure that no post-processing could repair, a *missing* sector, does not occur for either loss.

stage k	1	2	3	4	5	6	7	8
t_k ($B = 10^3$)	0.200	0.396	0.530	0.662	0.753	0.841	0.952	1.000
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.602	0.524	0.559	0.401	0.500	0.453	0.369	0.778
forward KL	0.391	0.398	0.463	0.366	0.467	0.465	0.400	0.708
t_k ($B = 10^4$)	0.200	0.396	0.588	0.720	0.849	1.000	—	—
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.703	0.628	0.381	0.412	0.336	0.303	—	—
forward KL	0.531	0.523	0.347	0.396	0.349	0.341	—	—
t_k ($B = 10^5$)	0.200	0.396	0.588	0.720	0.849	1.000	—	—
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.745	0.688	0.451	0.489	0.405	0.383	—	—
forward KL	0.611	0.602	0.416	0.461	0.400	0.387	—	—

Table 5: Validation ESS on every accepted rung, the acceptance gate of Algorithm 4 with floor $\tau_v = 0.3$, on the $p = 6$ clock model at $D = 64$; bold marks the better loss on each rung. Within each pair the two losses traverse the same accepted ladder and reject the same rungs, seven rejections per run at $B = 10^3$ and two at $B = 10^4$ and 10^5 , so the rows are directly comparable. The X-regularized loss leads on the early rungs; the gap closes in the transition window $t \approx 0.66$ – 0.95 , where the selector halves the increment and the sampling physics, shared by both losses, sets the ceiling. Wall time per pair is equal within 3%.

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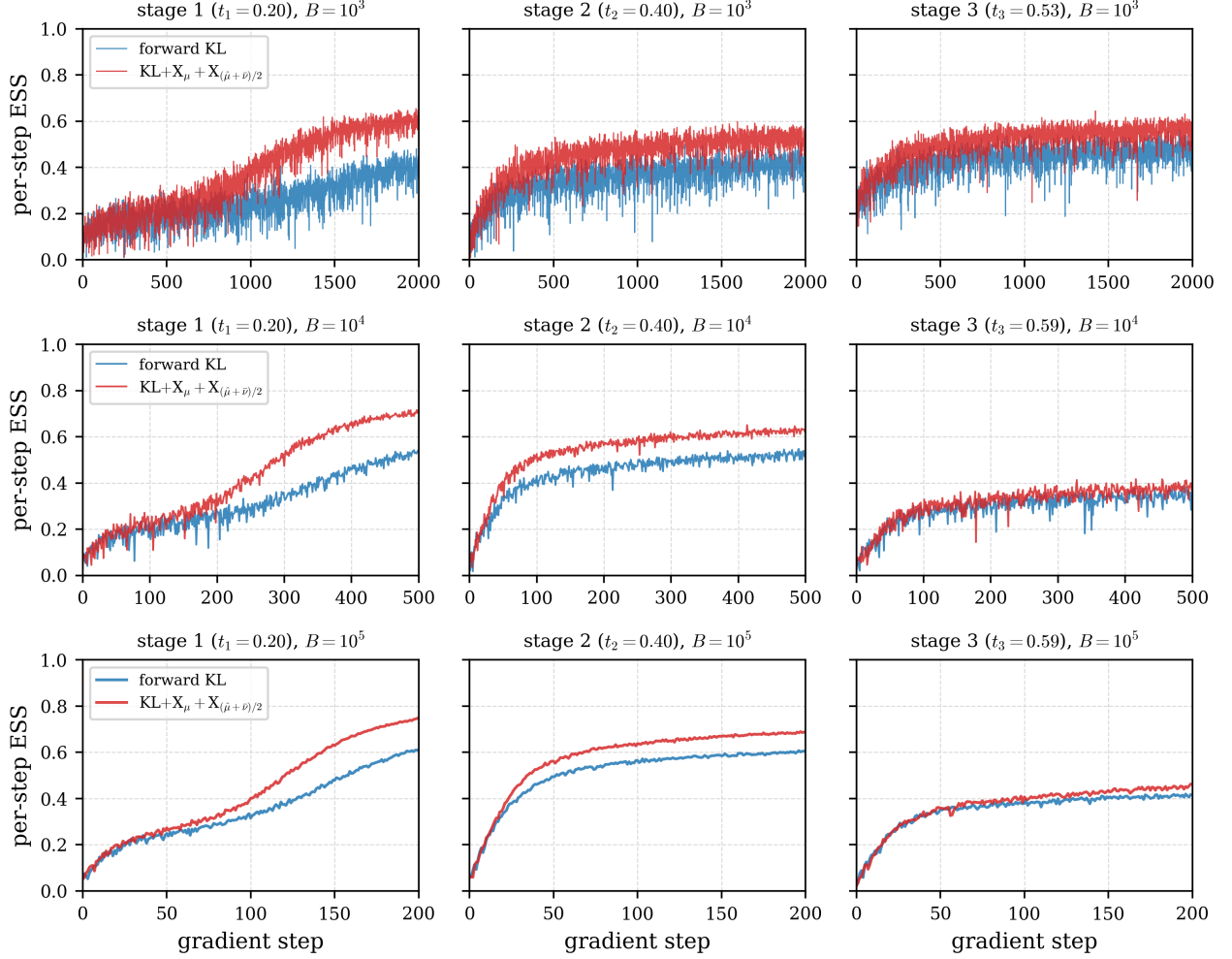


Figure 8: Per-step ESS of the training surrogate on the first three stages, bare forward KL (blue) versus the X-regularized loss (red); rows $B = 10^3, 10^4, 10^5$, one panel per stage. These are the far-but-smooth early bridges where the training loss is the binding constraint: the X-regularized curve plateaus visibly higher in every panel, mirroring the accepted-rung gates of Table 5.

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